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REVIEW

Recent Advances on the Chemical Composition, Pharmacological Properties, and Product Development of *Morus alba*

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ABSTRACT

Introduction: *Morus alba*, a member of the Moraceae family, has a long history of medicinal use in China. Its branches, fruits, leaves, and root bark are rich in various bioactive compounds, including flavonoids, alkaloids, and other phytochemicals. These plant parts exhibit a broad spectrum of pharmacological activities, such as antidiabetic, antitumor, antibacterial, and antioxidant effects. Owing to these properties, various parts of *M. alba* have been developed into functional foods and health products, attracting considerable public and scientific interest. **Objectives:** This review aims to summarize current knowledge on the phytochemistry, pharmacological activities, and product development related to different parts of *M. alba*, with particular emphasis on their respective roles in the development of functional products. **Methods:** Relevant literature was retrieved from major scientific databases, including PubMed, Web of Science, ScienceDirect, and Google Scholar. Information related to the phytochemistry, pharmacological effects, and product development of various parts of *M. alba* was systematically collected and analyzed. **Results:** This review summarizes the chemical composition of different parts of *M. alba* and their corresponding pharmacological properties. It also highlights recent advancements in the development of functional products derived from *M. alba*. **Conclusions:** *M.orus alba* contains a wide array of bioactive compounds with promising pharmacological potential. Its various parts have demonstrated potential in the prevention and management of chronic diseases and have been increasingly utilized in the development of functional health products. Further interdisciplinary studies are warranted to facilitate its translational application in modern healthcare.

1 | Introduction

Morus alba, a member of the Moraceae family, is a perennial deciduous tree or shrub widely distributed across Asia, Africa, Europe, Oceania, and other regions, with particular prevalence in Asia [1]. As one of the earliest countries to cultivate *M. alba*, China has a long history of *M. alba* cultivation. Furthermore, China is home to one of the richest diversities of

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Morus species, comprising 15 species and 4 varieties. Among them, black, red, and white mulberries are the three most widely distributed and dominant species globally [2]. Since ancient times, *M. alba* has not only served as a primary feed source for silkworms but has also been valued for the medicinal properties of its various parts. Specifically, the branches, fruits, leaves, root bark, and parasitic fungus are, respectively, utilized as medicinal materials known as *Mori Ramulus*, *Mori Fructus*, *Mori Folium*, *Mori Cortex*, and *Sanghuang*. This comprehensive utility has given rise to the saying, "*Morus alba* is a treasure trove." Today, *Mori Ramulus*, *Mori Fructus*, and *Mori Folium* are recorded in the *Pharmacopoeia of the People's Republic of China* and are widely used in traditional Chinese medicine (TCM). Among them, *Mori Fructus* and *Mori Folium* are recognized for their dual roles as both food and medicine. Modern studies have demonstrated that the medicinal parts of *M. alba* contain a wide range of bioactive compounds, including flavonoids, alkaloids, stilbenes, polyphenols, polysaccharides, amino acids, vitamins, and minerals [3–5]. The different parts of the *M. alba* exhibit a variety of pharmacological activities, including hypotensive, hypoglycemic, antineoplastic, antibacterial, and antioxidant effects [6]. With the continuous advancement of modern technology, our understanding of *M. alba* has significantly deepened, leading to substantial progress in the development and application of its components. Various proprietary health products and pharmaceutical formulations derived from *M. alba* have been successfully developed and have gained widespread recognition and acceptance. Simultaneously, the applications of *M. alba* components have expanded into the cosmetics and animal husbandry industries. This expansion has not only broadened the range of applications and extended the associated industrial chain but also facilitated a more comprehensive utilization of *M. alba* resources. This review provides a systematic summary of the bioactive constituents, pharmacological properties, and product developments derived from *M. alba*, offering crucial insights for a deeper understanding of its biological activities and the potential for further development and utilization of its medicinal resources.

2 | Medicinal Value of Various Parts of the *M. alba*

2.1 | Mori Folium

Mulberry leaves, the leaves of the *M. alba*, possess bitter, sweet, and cold in nature, entering the lung and liver meridians. They possess the ability to dispel wind-heat, clear the lungs, and moisten dryness, thereby clearing the liver and brightening the eyes. Mulberry leaves are commonly utilized in treating wind-heat colds, lung heat, dry cough, dizziness, headache, redness, and dimness of the eyes. According to the *Compendium of Materia Medica*, mulberry leaves are also recognized for their ability to quench thirst, promote diuresis, benefit the five viscera, unblock the joints, disperse stagnation, alleviate pain, and invigorate blood circulation. Hence, mulberry leaves are often referred to as "fairy grass" or "Tie shan zi." Mulberry leaves harvested after frost, also known as "frost mulberry leaves" or "winter mulberry leaves," are considered to have enhanced medicinal effects [7]. Renowned scholar Zhang Shouyi emphasized that mulberry leaves are most potent when they are old and have been exposed to frost; thus, winter mulberry leaves are specifically used for medicinal purposes. However, mulberry leaves from other seasons can also be utilized for treating conditions such as anal prolapse and spontaneous sweating. In addition to their medicinal uses, mulberry leaves can be consumed as a dietary component.

Modern research has demonstrated that mulberry leaves possess pharmacological effects, including hypoglycemic, hypotensive, antiviral, antibacterial, and antitumor properties. Consequently, they are not only widely used in preventive healthcare but are also regarded as "one of the top ten health foods for mankind in the 21st century" [8].

2.2 | Mori Ramulus

Mulberry twig, the dried branch of *M. alba* L., possesses a slightly bitter and neutral nature, primarily affecting the liver meridian. It is recognized for its ability to dispel rheumatism and promote joint mobility. Mulberry twigs are commonly used in treating rheumatic paralysis, as well as soreness and numbness in the shoulders, arms, and joints. According to the *Essentials of Materia Medica*, mulberry twigs "benefit the joints, nourish body fluids, move water to remove wind, and are particularly effective in treating numbness and pain in the upper limbs." The *Ben Cao Tu Jing* further records that mulberry twigs can treat wind, itching, and dryness throughout the body, as well as dry mouth. Traditionally, mulberry twigs were harvested in late spring and early summer, boiled in water for external use, or made into a paste for internal consumption. Ge Hong, a renowned physician, believed that "Mulberry twigs could be frequently used as they are neither cold nor hot in nature" [9].

Contemporary research has revealed that mulberry twigs possess a wide range of pharmacological properties, including anti-inflammatory, analgesic, hypoglycemic, antioxidant, and lipid-lowering effects. Clinically, mulberry twigs have been extensively employed in treating conditions such as soreness and numbness in the shoulders, arms, joints, hands, and feet; rheumatic paralysis; and various other disorders [10].

2.3 | Mori Fructus

Mulberries, the mature fruits of *M. alba*, possess a sweet, sour, and cold nature and primarily affect the heart, liver, and kidney meridians. According to the *Compendium of Materia Medica*, "Consuming mulberry juice alleviates alcohol poisoning, promotes diuresis, reduces swelling, nourishes yin, and replenishes blood. It is beneficial for liver and kidney deficiencies, essence and blood deficiencies, dizziness, blurred vision, tinnitus, insomnia, and premature graying of hair and beard." Mulberries are not only used as a medicinal remedy but also as a fruit [11]. They are colloquially referred to as the "Chinese fruit king." Modern research has confirmed that mulberries are rich in various nutrients and exhibit a range of pharmacological effects, including immunomodulatory properties, promotion of hematopoietic cell growth, as well as antineoplastic, hypoglycemic, antioxidant, antibacterial, and anti-inflammatory effects. In TCM, mulberries are widely used to treat joint disorders, lower blood pressure, protect against liver damage, and promote urination, among other applications.

2.4 | Mori Cortex

Sangbaipi, the dried root bark of *M. alba* with the cork bark removed, possesses a sweet and cold nature, entering the lung meridian. It is believed to have the effects of clearing lung heat to alleviate asthma and promoting diuresis to reduce edema. Sangbaipi is commonly used in treating lung fever, asthma, cough, edema, abdominal distention, urinary difficulties, and facial and skin swelling. According to the *Compendium of Materia Medica*, Sangbaipi specializes in promoting diuresis and is particularly effective for conditions involving excess fluid in the lungs and lung heat. In addition to its traditional pharmacological effects of suppressing cough, expectorating, and treating asthma, clinical studies have shown that Sangbaipi exhibits a diverse range of pharmacological properties, including anti-inflammatory, anticancer, antihypertensive, and hypoglycemic effects.

Sangbaipi has been demonstrated to be particularly effective in treating diseases in elderly individuals. For example, when combined with Western medicine, it can effectively manage acute exacerbations of chronic obstructive pulmonary disease (COPD), without compromising lung function [12]. Additionally, Sangbaipi decoctions have been found to significantly improve clinical symptoms in elderly COPD patients with phlegm-heat lung obstruction during acute exacerbations [13]. Furthermore, an herbal formula comprising mulberry leaves, twigs, and Sangbaipi, designed to clear heat, lower blood sugar, and promote blood circulation, has proven effective in treating and preventing diabetic peripheral neuropathy with hyperglycemia in elderly individuals.

2.5 | Sanghuang

Sanghuang, also known as the "sang er" and "sang chen," is a medicinal fungus parasitized on *M. alba* [14]. It was first mentioned in *The Characters of Drugs* during the Tang Dynasty, indicating its use in China for over 2000 years, and has been documented in various TCM classics [15, 16]. The *Shennong Materia Medica Classic* records that Sanghuang can lighten the body and prolong life, while *The Compendium of Materia Medica* states that it can treat conditions such as hematochezia, hemorrhagia, and amenorrhea. Additionally, *The Characters of Drugs* from the Tang Dynasty records its use for hemorrhage, hemostasis, and female menstruation-related disorders. Based on the synthesis of these classical texts, Sanghuang has demonstrated significant therapeutic efficacy in the treatment of gynecological disorders and deficiency-related conditions, thereby earning it the reputation of "forest gold." Modern pharmacological research has revealed that Sanghuang contains a diverse array of bioactive compounds. Not only can it enhance human immunity, but its antioxidant, lipid-lowering, hepatoprotective, and antineoplastic effects have also demonstrated significant therapeutic potential [17, 18].

3 | Phytoconstituents of *M. alba*

Contemporary research has revealed that various parts of *M. alba* contain a wide range of bioactive compounds, which provide valuable insights for the rational development and utilization of *M. alba* in medicine and food. This section summarizes the chemical composition of *M. alba* according to seven structural types, including flavonoids, alkaloids, stilbenoids, Diels–Alder adducts, and terpenes [1, 19].

3.1 | Polysaccharide Compounds

Polysaccharides in *M. alba* are abundant and exhibit considerable structural diversity across different plant parts, comprising various monosaccharide units linked by complex glycosidic bonds. According to previous studies, these polysaccharides are primarily composed of mannose, galactose, glucose, arabinose, galacturonic acid, and glucuronic acid, along with trace amounts of sorbitol, ribose, and fucose [20]. Variations in monosaccharide composition and molar ratios may be attributed to differences in processing, extraction, and purification methods. The polysaccharides in *Sanghuang* are primarily acidic heteropolysaccharides, predominantly composed of glucose, with mannose present in the lowest proportion [21]. Carbohydrates in mulberries exhibit a distinct distribution pattern, with glucose content approximately eight times higher than that of sucrose and fructose about five times higher [22]. Notably, mulberries contain the functional polysaccharide inulin, which has been associated with significant physiological activities, including blood glucose regulation, gut health promotion, and weight management [23]. In addition to their nutritional roles, mulberry polysaccharides play a key role in plant responses to environmental stress. For example, under flooding stress, endogenous polysaccharides are converted into soluble sugars to maintain physiological homeostasis [24]. Polysaccharides from *M. alba* also exhibit a wide range of biological activities, including antidiabetic, antioxidant, immunomodulatory, and hepatorenal protective effects. Based on current literature, this review systematically summarizes the polysaccharide composition and distribution characteristics across different parts of *M. alba*, with those from mulberry fruits and leaves accounting for the majority, as detailed in Table 1.

3.2 | Flavonoids Compounds

Flavonoids are among the primary physiologically active compounds found in *M. alba*. They possess a basic C_6–C_3–C_6 structural skeleton, comprising two benzene rings, denoted as A and B, interconnected via a heterocyclic ring C. Hydroxyl groups are frequently substituted at the C_5, C_7, and $C_{4'}$ positions [50]. The structural diversity of flavonoids arises from varying degrees of oxidation and modifications to the heterocyclic ring C, resulting in the formation of distinct flavonoid classes [51]. Flavonoids also play diverse and important roles in plant physiology. For example, they are essential for drought resistance in *M. alba*, enhancing tolerance to drought stress by reducing the accumulation of reactive oxygen species (ROS) [52]. Anthocyanins, a major subclass of flavonoids, are responsible for the red, purple, and other vibrant colors of mulberry fruits. In addition to their physiological roles in plants,

TABLE 1 | The detailed information on polysaccharide compounds in different parts of *Morus alba*.

Components	Monosaccharide composition	Sources	Reference
CMA-a-1	Glc	Mori Cortex (root bark)	[21]
CMA-a-5	Glc	Mori Cortex (root bark)	[21]
CMA-b1-1	Rha: Ara: Xyl: Gal: GalA: Glc (8.6: 31.3: 11.0: 24.6: 16.9: 7.4)	Mori Cortex (root bark)	[21]
BMP-1-1	Man: Rha: GlcA: Gal: Ara (1.46: 2.14: 2.16: 48.27: 45.97)	Mori Fructus	[25]
BMP-2-1	Rha: GlcA: GalA: Gal: Ara (23.55: 2.04: 27.5: 22.38: 24.54)	Mori Fructus	[25]
FMP-6-S2	Rha: GalA: Gal: Ara (30.91: 24.78: 28.70: 15.61)	Mori Fructus	[26]
MFP-II	GalA: Glc: Gal: Man: Rha: Xyl: Ara (3.6: 23.1: 10.9: 2.9: 30.5: 2.7: 26.3)	Mori Fructus	[27]
P1-MFPs	Man: Rha: GlcA: Glc: Xyl (11.0: 5.4: 4.7: 33.5: 45.5)	Mori Fructus	[28]
MFP-EAHE100	Man: Rha: GalA: Glc: Gal: Ara: Fuc (2.57: 1.24: 0.19: 5.24: 1.00: 0.51: 1.06)	Mori Fructus	[29]
MFP-UAHE100	Man: Rha: GalA: Glc: Gal: Ara: Fuc (3.47: 1.45: 0.21: 6.96: 1.00: 0.47: 1.55)	Mori Fructus	[30]
MFP-1	Gal: Glc: Ara: Rha: Man (26.80: 20.40: 8.70: 5.00: 1.00)	Mori Fructus	[30]
MFP-2	Ara: Gal: Rha: Glc: GalA (34.2: 38.2: 8: 17.5: 15.1)	Mori Fructus	[30]
MFP-3	Glc: Gal: Rha: GalA: Ara: GlcA (47.4: 34.9: 36.1: 33.1: 19.9: 4.1)	Mori Fructus	[30]
MFP-4	Ara: Gal: Glc: GalA (59.86: 27.15: 12.99: 31.90)	Mori Fructus	[30]
FPA-1	Rha: GalA: Gal: Ara (1.0: 2.4: 2.7:4.9)	Mori Fructus	[31]
FPA-2	GalA: Gal: Ara (36.93: 11.19: 19.69)	Mori Fructus	[31]
FPA-4	Rha: GalA: Ara (14.6: 38.04: 15.8)	Mori Fructus	[31]
FPA-5	Rha: GalA: Ara (5.81: 44.38: 21.66)	Mori Fructus	[31]
MPS-1	Sor: Ara: Xyl: Glu (3: 2: 17: 110)	Mori Folium	[32]
MPS-2	Rha: Ara: Xyl: Glu: Gal: Man (13.8: 9.7: 14.4: 9.6: 9.3: 13.5)	Mori Folium	[32]
MLP-2	Rha: Ara: Xyl: Man: Gal: Glc (7.36: 10.26: 2.25: 0.43: 2.51: 1.00)	Mori Folium	[33]
Had-1	Ara: Man: Rha: Xyl: GlcA: Glu: Gal (0.76: 1.23: 0.5: 0.33: 0.32: 1: 1)	Mori Folium	[34]
HAD-2	Ara: Man: Rha: Xyl: GlcA: Glu: Gal (2.74: 1.5: 0.56:1.43: 0.21: 1: 3.28)	Mori Folium	[34]
HAD-3	Ara: Man: Rha: Xyl: GlcA: Glu: Gal (2.74: 1.5: 0.56: 1.43: 0.21:1: 3.28)	Mori Folium	[34]
AD	Ara: Man: Rha: Xyl: GlcA: Glu: Gal (1.72: 1.25: 0.44: 0.41: 0.28: 1: 2.34)	Mori Folium	[34]
SDA	Rha: Ara: Xyl: Glu: Gal: GalA (5: 4: 1: 2: 6: 38)	Mori Folium	[35]
MLP-2C	Rha: Ara: Xyl: Man: Glu: Gal: GluA: GalA (1.31: 0.09: 3.56: 20.88: 34.39: 4.81: 21.71: 13.25)	Mori Folium	[36]
PIP60–1	Fuc: Glc: Man: Gal: MeGal (1: 1: 1: 2: 1)	Phellinus	[37]
P1	Fru: Rha: Gal: Glc: Xyl: Man: MeGal (1: 3.12: 33.51: 2.0: 4.03: 1.09: 2.87)	Phellinus	[38]
PIP-1	Man: Glc: Gal (2.41: 87.74: 3.86)	Phellinus	[39]
PIE	Xyl: Man: Fuc: Glc: Gal (2.3: 1: 6.4: 22.1: 19.83)	Phellinus	[40]
PIPS	Glc: rhamnose: Man (11.0: 14.0: 1.0)	Phellinus	[41]
FIII-1	Fuc: Xyl: Man: Glc: Gal (1.1: 3.6: 3.4: 87.3: 4.7)	Phellinus	[42]
PLPS-1	Glc: Ara: Fuc: Gal: Xyl (21.964: 1.336: 1.182: 1: 1)	Phellinus	[43]

(Continues)

TABLE 1 | (Continued)

Components	Monosaccharide composition	Sources	Reference
PLPS-2	Gal: Glc: Man: Ara: Fuc: Xyl (14.368: 2.594: 1.956: 1.552: 1.466: 1)	Phellinus	[43]
PL-N1	Ara: Xyl: Glc: Gal (4.0: 6.7: 1.3: 1.0)	Phellinus	[44]
PL-A11	Ara: Xyl: Man: Glc (1.1: 1.3: 1.0: 6.6)	Phellinus	[45]
RMP1	Ara: Xyl: Glc: Gal: Rha (0.56: 0.37: 0.17: 1: 0.08)	Mori Ramulus	[46]
RMP	Man: Rha: GlcA: Glc: Xyl: Gal: Ara (1.36: 2.68: 0.46: 328.17: 1.53: 21.80: 6.16)	Mori Ramulus	[47]
RMP	Glc: Gal: Man: Rha: Ara (62.1: 13.8: 6.7: 11.6: 5.8)	Mori Ramulus	[48]
MTP30	Rha: Ara: Gal: Glc: Xyl: Man: GalA (2.10: 1.62: 1.95: 86.80: 0.18: 0.36: 6.73)	Mori Ramulus	[49]
MTP60	Rha: Ara: Gal: Glc: Xyl: Man: GalA (5.53: 5.49: 8.85: 62.90: 1.05: 1.39: 13.6)	Mori Ramulus	[49]
MTP90	Rha: Ara: Gal: Glc: Xyl: Man: GalA (1.05: 7.42: 11.8: 69.00: 1.79: 5.60: 2.55)	Mori Ramulus	[49]

flavonoids are the predominant class of active phytochemicals in *M. alba*, exerting multiple pharmacological effects in animals. These include antioxidant, anti-inflammatory, cardioprotective, antidiabetic, and anti-obesity activities. The flavonoid components and source parts of various parts of the *M. alba* are shown in Table 2, and the chemical structure is shown in Figure 1. The biosynthetic pathway of key flavonoid compounds in *M. alba* is shown in Figure 2.

3.3 | Alkaloids Compounds

Alkaloids are widely distributed across various parts of *M. alba*. Among these, 1-deoxynojirimycin (DNJ) is a prominent alkaloid found in mulberry and has attracted significant attention and extensive research due to its crucial biological activities. DNJ, a polyhydroxy alkaloid containing an imine group, is known for its strong insecticidal activity and plays a key role in protecting *M. alba* against herbivory [83]. The distribution of DNJ among the total alkaloids in different parts of the *M. alba* follows the order: sangbaipi > mulberry branches > mulberry leaves [84]. However, the alkaloid content in *M. alba* is not constant and is influenced by factors such as medicinal parts, varieties, environmental conditions, and growth and maturation stages [85]. For example, the DNJ content in mulberry leaves peaks at the bud stage and then decreases as the leaves mature, following the order: young shoots > young leaves > mature leaves > old leaves [86]. In mulberry branches, DNJ content varies significantly with the growth stage, increasing progressively as the branches mature, peaking at around 4 months before gradually declining [87]. At equivalent monthly ages, DNJ content in mulberries decreases as they mature, with significant differences observed between varieties [82, 88]. The chemical composition and sources of alkaloids in various parts of the *M. alba* are shown in Table 3, and the chemical structure is shown in Figure 3. The biosynthetic pathway of key alkaloid compounds in *M. alba* is shown in Figure 4.

3.4 | Other Types of Compounds

In addition to the previously mentioned compounds, *M. alba* also contains other important chemical components, primarily stilbenes, terpenes, coumarins, and anthocyanins. Among these, stilbene compounds refer to a class of compounds polymerized with resveratrol or other stilbenes as monomers. They are classified into three categories: stilbenes, 2-phenylbenzofurans, and stilbene oligomers. The chemical composition and sources of terpenoids, coumarins, benzofurans, and stilbenes in various parts of *M. alba* are summarized in Table 4, with their structures depicted in Figure 5a–d. The biosynthetic pathway of moracins in *M. alba* is illustrated in Figure 6.

4 | Pharmacological Activities of *M. alba*

Various parts of the *M. alba*, including the root bark, leaves, and fruits, have a long history of use in TCM and folk remedies. These parts contain a diverse range of monomeric and polymeric bioactive compounds, which demonstrate notable therapeutic effects against various diseases. This article examines the traditional medicinal properties of the different parts of *M. alba* and integrates modern pharmacological findings to elucidate their mechanisms of action.

4.1 | Antidiabetic Activity

Diabetes mellitus (DM) is the third most prevalent chronic progressive disease worldwide, following cardiovascular diseases and cancer. Clinically, diabetes is characterized by symptoms such as polydipsia, polyphagia, polyuria, and weight loss [100]. Although conventional hypoglycemic agents are effective in reducing blood glucose levels, their clinical utility is often constrained by limitations such as drug resistance, hypoglycemia risk, and potential hepatic and renal toxicity [100]. Therefore, the

TABLE 2 | The detailed information on flavonoid compounds in different parts of *Morus alba*.

No.	Classifications	Components	Sources	Reference
1	Flavonoids	3′-Geranyl-3-prenyl-2′,4′,5,7-tetrahydroxyflavone	Mori Folium	[53]
2		3′,8-Diprenyl-4′,5,7-trihydroxy flavanone	Mori Folium	[53]
3		8-Geranylapigenin	Mori Folium	[53]
4		Kuwanon S	Mori Folium	[53]
5		Kuwanon C	Mori Cortex (root bark) Mori Ramulus	[54, 55]
6		Kuwanon T	Mori Cortex (root bark)	[55, 56]
7		Albanin D	Mori Ramulus	[57]
8		5′-Geranyl-5,7,2′,4′-tetrahydroxyflavone	Mori Cortex (root bark)	[56]
9		Albanin A	Mori Cortex (root bark)	[56]
10		5′-(1″,1″-Dimethylallyl)-5,7,2′,4′-tetrahydroxyflavone	Mori Cortex (root bark)	[56]
11		Norartocarpetin	Mori Folium	[58]
12		Cudraflavone C	Mori Cortex (root bark)	[56]
13		Sanggenon K	Mori Folium	[53]
14		Kuwanon A	Mori Cortex (root bark)	[55]
15		Morusin	Mori Folium Mori Cortex (root bark)	[53, 55, 56]
16		Atalantoflavone	Mori Folium	[53]
17		Morusinol	Mori Cortex (root bark) Mori Ramulus	[54, 57]
18		Sanggenon J	Mori Folium	[53]
19		Kuwanon B	Mori Cortex (root bark)	[59]
20		Cudraflavone B	Mori Cortex (root bark)	[54]
21		Cyclomulberrin	Mori Cortex (root bark) Mori Ramulus	[53, 57]
22		Cyclomorusin	Mori Folium	[53]
23		(2*S*)-7,4′-Dihydroxyflavanone	Mori Fructus	[60]
24		Eriodictyol	Mori Fructus	[60]
25		(2*R*,3*R*)-Dihydroquercetin	Mori Fructus	[60]
26		Albanin C	Mori Cortex (root bark)	[56]
27		Albanin T	Mori Cortex (root bark)	[56]
28		Mulberranol	Mori Ramulus	[57]
29	Flavonols	Isoquercitrin	Mori Folium	[61, 62]
30		Kaempferol	Mori Fructus	[53, 60]
31		Kaempferol 3-*O*-β-D-glucopyranoside	Mori Fructus	[63]
32		Kaempferol 3-*O*-β-D-rutinoside	Mori Fructus	[63]

(Continues)

TABLE 2 | (Continued)

No.	Classifications	Components	Sources	Reference
33		Quercetin	Mori Fructus Mori Folium Mori Ramulus	[58, 60, 63, 64]
34		Quercetin 3-*O*-β-D-glucopyranoside	Mori Fructus	[63, 65]
35		Quercetin 3-0-(6″-0-acetyl)-8-D-glucopyranoside	Mori Fructus	[63]
36		Quercetin 3-*O*-β-D-rutinoside	Mori Fructus	[63, 65]
37		Quercetin 7-*O*-β-D-glucopyranoside	Mori Fructus	[63]
38		Quercetin 3,7-di-*O*-β-D-glucopyranoside	Mori Fructus	[63]
39		5,7,2′,4′-Tetrahydroxy-3-methoxyflavone	Mori Ramulus	[57]
40		Morin	Mori Ramulus	[64]
41		Astragalin	Mori Folium	[62]
42	Dihydroflavone	Kuwanon E	Mori Cortex (root bark)	[54, 55, 66]
43		Sanggenol Q	Mori Cortex (root bark)	[56, 67]
44		Kuwanon U	Mori Cortex (root bark)	[56]
45		5,7,3′-Trihydroxyflavanone-4′-*O*-β-D-glucopyranoside	Mori Fructus	[63]
46		5,7,4′-Trihydroxyflavanone-3′-*O*-β-D-glucopyranoside	Mori Fructus	[63]
47		Naringenin	Mori Fructus	[60]
48		Eriodictyol	Mori Ramulus	[57]
49		Steppogenin	Mori Ramulus	[57]
50		Sanggenon H	Mori Cortex (root bark)	[54, 66]
51		Sanggenol L	Mori Cortex (root bark)	[56]
52		Sanggenol O	Mori Cortex (root bark)	[56, 68]
53		Benzokuwanon E	Mori Cortex (root bark)	[66]
54		Sanggenon F	Mori Cortex (root bark)	[55]
55		Kuwanon F	Mori Cortex (root bark)	[66]
56	Dihydroflavonol	Dihydrokaempferol 7-*O*-β-D-glucopyranoside	Mori Fructus	[63]
57		Dihydrokaempferol	Mori Ramulus	[57]
58		Dihydromorin	Mori Ramulus	[69]
59		Dihydromorin 7-*O*-β-glucopyranoside	Mori Ramulus	[69]
60	Chalcones	Isobavachalcone	Mori Cortex (root bark) Mori Fructus	[60, 63]
61		Morachalcone A	Mori Folium Mori Ramulus	[62, 66, 70]
62		Morachalcone D	Mori Folium	[62]

(Continues)

No.	Classifications	Components	Sources	Reference
63		2,4,2′,4′-Tetrahydroxychalcone	Mori Ramulus	[57]
64		Morachalcone C	Mori Folium	[70]
65		Morachalcone B	Mori Folium	[55]
66		(2*E*)-1-[2,3-Dihydro-4-hydroxy-2-(1-methylethenyl)-5-benzofuranyl]-3-(4hydroxyphenyl)-1-propanone	Mori Fructus	[63]
67		Morachalcone E	Mori Folium	[62]
68	Diels–Alder-type adduct	Kuwanon H	Mori Cortex (root bark)	[59]
69		Kuwanon G	Mori Fructus Mori Cortex (root bark)	[71, 72]
70		Sanggenon C	Mori Fructus	[71, 73]
71		Sanggenon E	Mori Cortex (root bark)	[54]
72		Soroceal	Mori Cortex (root bark)	[54]
73		Albafuran C	Mori Ramulus	[74]
74		Sorocein A	Mori Ramulus	[74]
75		Mulberrofuran E	Mori Ramulus	[74]
76		Albasin B	Mori Cortex (root bark)	[56]
77		Mulberrofuran J	Mori Ramulus	[75]
78		Kuwanon J	Mori Ramulus	[59, 75]
79		Kuwanon I	Mori Cortex (root bark)	[59]
80		Mulberrofuran F_1	Mori Folium	[59]
81		Mulberrofuran F	Mori Ramulus Mori Folium	[59, 74]
82		Mulberrofuran G	Mori Cortex (root bark)	[56]
83		Morusalbin A	Mori Cortex (root bark)	[56]
84		Morusalbin B	Mori Cortex (root bark)	[56]
85		Morusalbin C	Mori Cortex (root bark)	[56]
86		Morusalbin D	Mori Cortex (root bark)	[56]
87		Macrourin G	Mori Cortex (root bark)	[56]
88		Yunanensin A	Mori Cortex (root bark)	[56]
89		Mulberrofuran K	Mori Cortex (root bark)	[56]
90		Sanggenons O	Mori Cortex (root bark)	[73]
91		Cathayanon A	Mori Cortex (root bark)	[73]
92		Cathayanon B	Mori Cortex (root bark)	[73]
93		Guangsangons G	Mori Ramulus	[75]
94		Guangsangons I	Mori Ramulus	[75]
95		Guangsangons F	Mori Ramulus	[75]
96		Guangsangons H	Mori Ramulus	[75]

(Continues)

TABLE 2 | (Continued)

No.	Classifications	Components	Sources	Reference
97		Mongolicin D	Mori Cortex (root bark) Mori Cortex (stem bark) Mori Ramulus	[76]
98		Kuwanon W	Mori Cortex (stem bark)	[77]
99		Mesozygin A	Mori Folium	[78]
100		Mesozygin B	Mori Folium	[78]
101		Mesozygin C	Mori Folium	[78]
102		Artonin I	Mori Folium	[78]
103	Anthocyanin	Cyanidin-3-*O*-hexoside	Mori Fructus	[79, 80]
104		Pelargonidin 3-*O*-hexoside	Mori Fructus	[79, 80]
105		Delphinidin acetyl-hexoside	Mori Fructus	[79, 80]
106		Cyanidin 3-*O*-glucoside	Mori Fructus	[65, 81]
107		Cyanidin 3-*O*-rutinoside	Mori Fructus	[65, 81]
108		Pelargonidin 3-*O*-glucoside	Mori Fructus	[82]
109		Pelargonidin 3-*O*-rutinoside	Mori Fructus	[82]
110		Cyanidin 3-*O*-(6″-malonyl-glucoside)	Mori Fructus	[81]

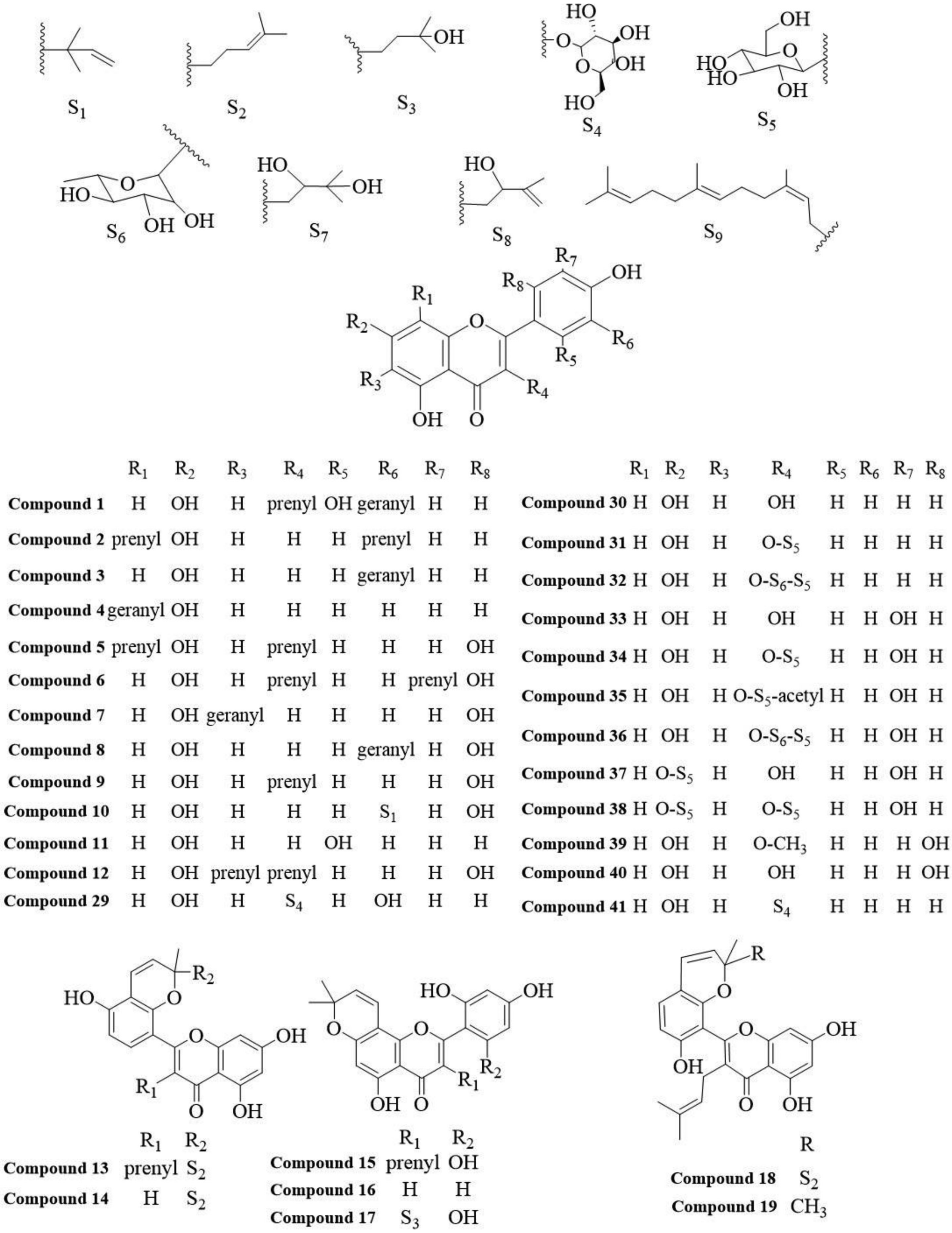
development of novel hypoglycemic agents with improved safety profiles and minimal adverse effects remains a pressing priority.

TCM and its bioactive compounds, with their multicomponent nature and long history of use, represent an important source of natural and safe hypoglycemic agents [101, 102]. Among these, DNJ, a key alkaloid predominantly found in mulberry leaves, plays a crucial role in maintaining glucose homeostasis and alleviating postprandial hyperglycemia. Studies have shown that DNJ specifically binds to the active site of α-glucosidase, thereby blocking the binding of disaccharides to the enzyme, which reduces the burden on pancreatic β-cells and exerts a marked hypoglycemic effect [103]. In vitro experiments further confirmed that DNJ extracted from mulberry leaves significantly promotes glucose uptake in differentiated 3T3-L1 adipocytes, thereby contributing to the regulation of glucose homeostasis [104]. Alkaloids isolated from mulberry twigs also exhibit potent α-glucosidase inhibitory activity [105]. Additionally, alkaloids isolated from mulberry twigs exhibit potent α-glucosidase inhibitory activity, indicating that despite originating from different plant parts, mulberry leaves and twigs share similar bioactive compounds and mechanisms of action. These findings collectively support the broad potential of mulberry-derived resources in the development of functional health products aimed at improving glucose metabolism and managing diabetes.

Flavonoids exert antidiabetic effects through multiple mechanisms, including preserving the structural integrity of pancreatic β-cells and enhancing insulin sensitivity. Kaempferol, a major flavonoid found in mulberry leaves, has been shown to act synergistically with gut microbiota to selectively inhibit the activity of PTP1B, suggesting therapeutic potential not only for Type 2 diabetes but also for Alzheimer's disease [106]. In addition, quercetin-3-*O*-glucoside and isorhamnetin present in mulberry leaves exhibit significant inhibitory activity against α-glucosidase [107]. Notably, flavonoids from mulberry fruits also demonstrate potent α-glucosidase inhibition. The synergistic interaction between DNJ and vitexin has been demonstrated to significantly enhance this inhibitory effect [108]. Furthermore, flavonoids derived from mulberry root bark exhibit beneficial effects on glucose metabolism, including the regulation of glucose homeostasis, enhancement of insulin sensitivity, and promotion of cellular glucose uptake [109].

Polysaccharides extracted from mulberry leaves have shown significant efficacy in improving insulin resistance and glucose metabolism disorders in Type 2 diabetic rat models, while also exerting clear protective effects against hepatic and renal tissue damage associated with diabetes. These multitarget regulatory effects are primarily mediated through the modulation of the IR and the TGF-β/Smads signaling pathway. The hypoglycemic activity of polysaccharides derived from mulberry twigs is attributed to their ability to inhibit pancreatic inflammation and oxidative stress, thereby improving peripheral insulin sensitivity and increasing serum insulin levels [48]. Furthermore, polysaccharides isolated from Sanghuang mushrooms exhibit potential inhibitory activity against both α-amylase and α-glucosidase [110]. The gut microbiota plays a critical role in the onset and progression of diabetes. Polysaccharides from mulberry fruits ameliorate symptoms such as weight loss, hyperglycemia, and insulin resistance in diabetic mice by modulating the composition and function of gut microbiota [111].

Overall, these findings highlight the therapeutic potential of different parts of the *M. alba* in managing diabetes-associated hyperglycemia and hyperlipidemia. Future research should focus



	R$_1$	R$_2$	R$_3$	R$_4$	R$_5$	R$_6$	R$_7$	R$_8$
Compound 1	H	OH	H	prenyl	OH	geranyl	H	H
Compound 2	prenyl	OH	H	H	H	prenyl	H	H
Compound 3	H	OH	H	H	H	geranyl	H	H
Compound 4	geranyl	OH	H	H	H	H	H	H
Compound 5	prenyl	OH	H	prenyl	H	H	H	OH
Compound 6	H	OH	H	prenyl	H	H	prenyl	OH
Compound 7	H	OH	geranyl	H	H	H	H	OH
Compound 8	H	OH	H	H	H	geranyl	H	OH
Compound 9	H	OH	H	prenyl	H	H	H	OH
Compound 10	H	OH	H	H	H	S$_1$	H	OH
Compound 11	H	OH	H	H	OH	H	H	H
Compound 12	H	OH	prenyl	prenyl	H	H	H	OH
Compound 29	H	OH	H	S$_4$	H	OH	H	H

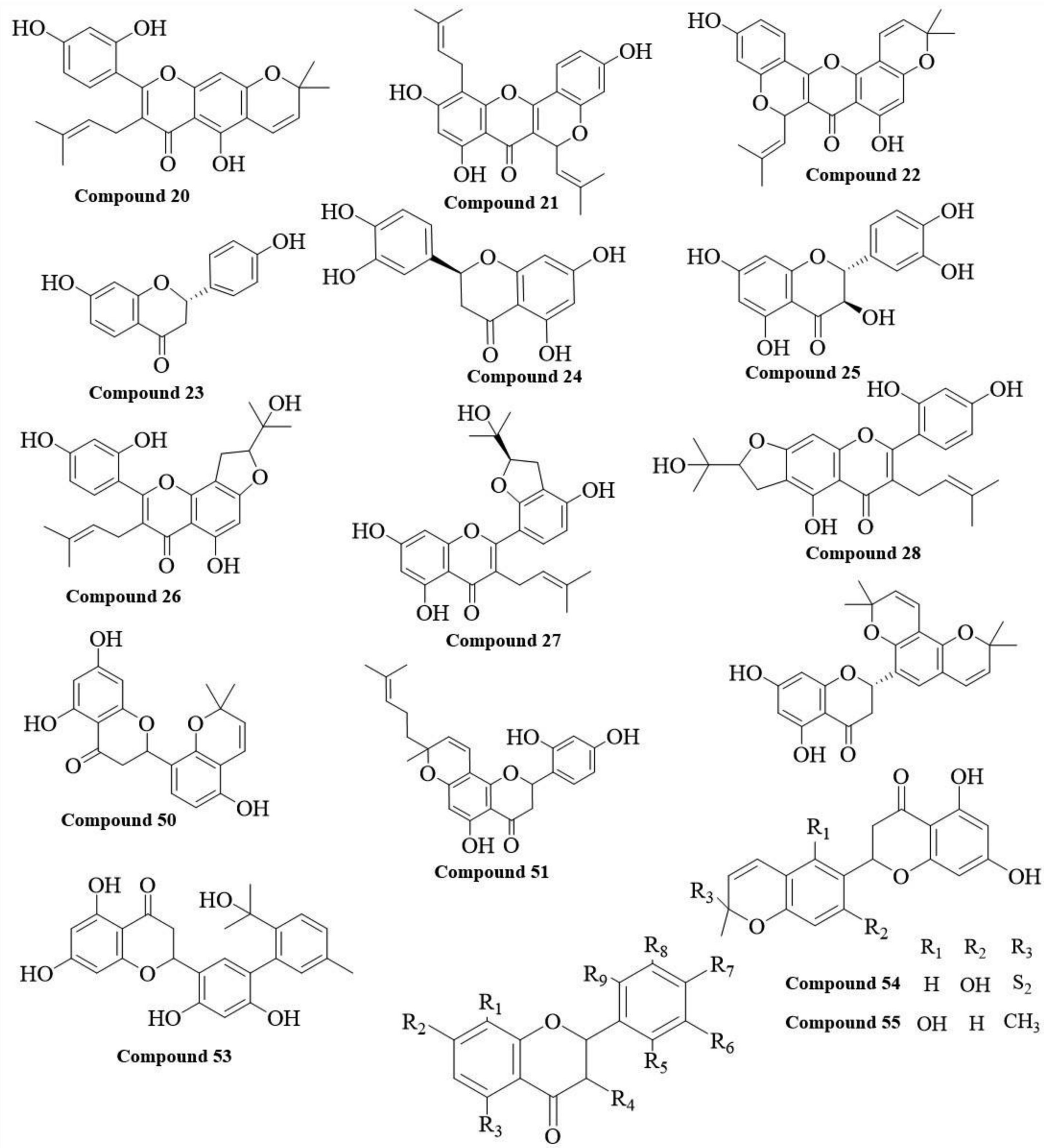
	R$_1$	R$_2$	R$_3$	R$_4$	R$_5$	R$_6$	R$_7$	R$_8$
Compound 30	H	OH	H	OH	H	H	H	H
Compound 31	H	OH	H	O-S$_5$	H	H	H	H
Compound 32	H	OH	H	O-S$_6$-S$_5$	H	H	H	H
Compound 33	H	OH	H	OH	H	H	OH	H
Compound 34	H	OH	H	O-S$_5$	H	H	OH	H
Compound 35	H	OH	H	O-S$_5$-acetyl	H	H	OH	H
Compound 36	H	OH	H	O-S$_6$-S$_5$	H	H	OH	H
Compound 37	H	O-S$_5$	H	OH	H	H	OH	H
Compound 38	H	O-S$_5$	H	O-S$_5$	H	H	OH	H
Compound 39	H	OH	H	O-CH$_3$	H	H	H	OH
Compound 40	H	OH	H	OH	H	H	H	OH
Compound 41	H	OH	H	S$_4$	H	H	H	H

FIGURE 1 | The flavonoids chemical structures in different parts of *Morus alba*.

on translating these experimental outcomes into pharmaceutical agents or functional health products aimed at the prevention and treatment of diabetes.

4.2 | Antioxidant Activity

The antioxidant levels of mulberries vary according to their color. Studies have demonstrated that purple mulberries contain higher anthocyanin levels than red mulberries, offering a notable advantage in both anthocyanin content and antioxidant capacity [112]. In comparison to the white mulberry genotype, the black mulberry genotype has a higher concentration of phenolic compounds, leading to enhanced antioxidant properties [22]. During the maturation of mulberries, the DPPH radical scavenging activity increases gradually. In contrast, mulberry leaves show the opposite trend, with younger leaves exhibiting significantly higher DPPH radical scavenging activity compared to mature leaves [113].



	R1	R2	R3	R4	R5	R6	R7	R8	R9
Compound 42	H	OH	OH	H	H	geranyl	OH	H	OH
Compound 43	H	OH	OH	H	H	prenyl	OH	prenyl	OH
Compound 44	H	OH	OH	H	H	geranyl	O-CH_3	H	OH
Compound 45	H	OH	OH	H	H	H	O-S_5	OH	H
Compound 46	H	OH	OH	H	H	H	OH	O-S_5	H
Compound 47	H	O-S5-S6	OH	H	H	H	OH	H	H
Compound 48	H	OH	OH	H	H	H	OH	OH	H
Compound 49	H	OH	OH	H	OH	H	OH	H	H
Compound 56	H	O-S_5	OH	OH	H	H	OH	H	H
Compound 57	H	OH	OH	OH	H	H	OH	H	H
Compound 58	H	OH	OH	OH	H	H	OH	H	OH
Compound 59	H	O-S_5	OH	OH	H	H	OH	H	OH

FIGURE 1 | (Continued)

Compared to quercetin and resveratrol, morusin N, extracted from mulberry leaves, showed weaker free radical scavenging activity. However, it displayed stronger DPPH radical scavenging activity and cell-based antioxidant effects [114]. Additionally, mulberry leaf polysaccharides demonstrated antioxidant activity, enhancing antioxidant capacity in a dose-dependent manner [115].

Sanghuang extract also exhibited strong antioxidant activity. Flavonoids and terpenoids in the extract demonstrated potent scavenging effects on DPPH, $ABTS^{\cdot+}$, and superoxide anion radicals [116, 117]. Additionally, mulberry fungus polysaccharides exhibited significant antioxidant activity [118].

Given the dual roles of mulberry fruit and leaves as both food and medicine, along with their pronounced antioxidant activities, they can be formulated in combination with other medicinal materials to develop a wide range of functional foods. This approach not only enhances the pharmacological efficacy and

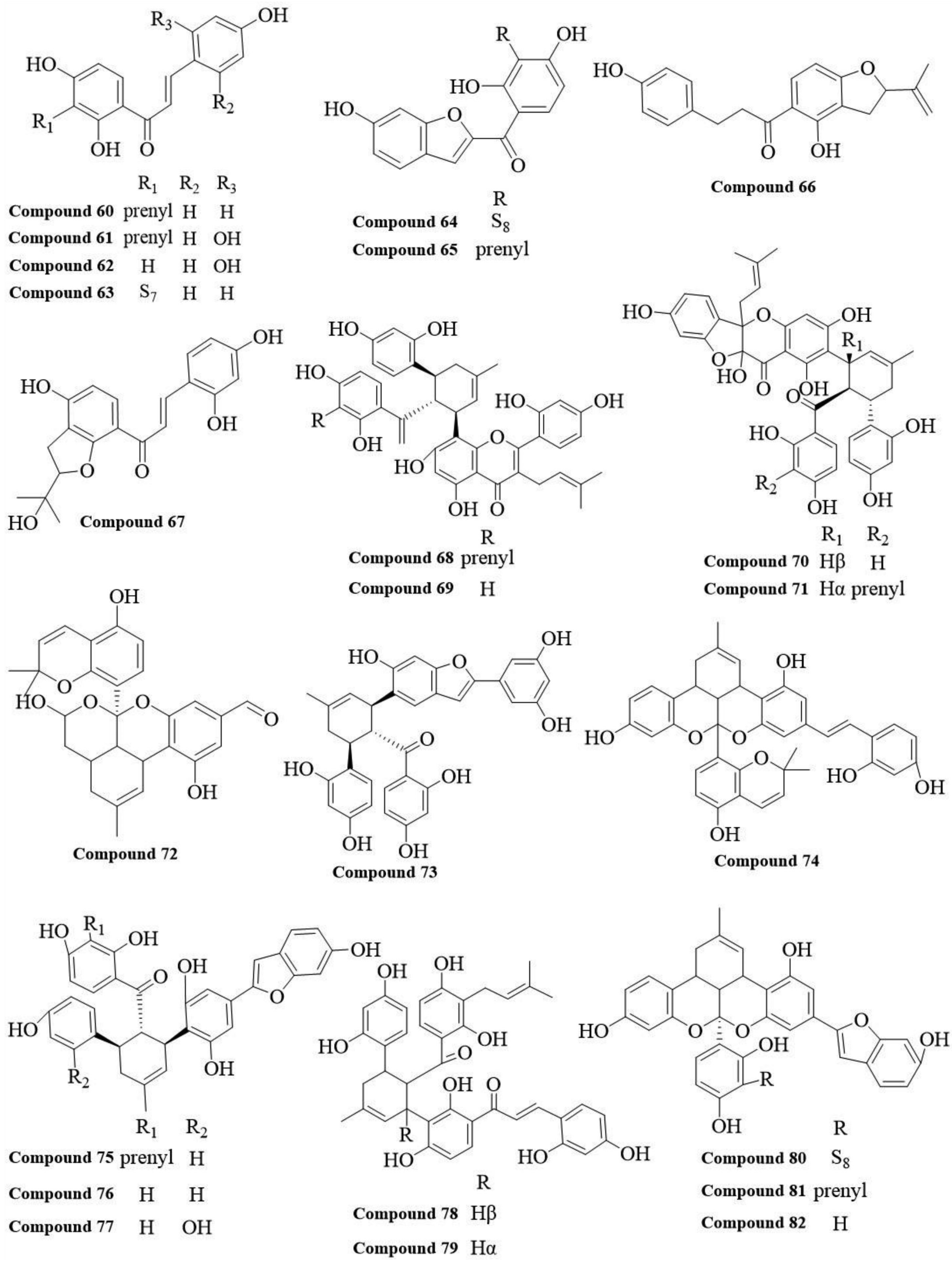


FIGURE 1 | (Continued)

functional value of the final products but also caters to consumer demand for highly effective antioxidant-based health solutions.

4.3 | Anti-Inflammatory Activity

The inflammatory response is a protective mechanism activated by the body in response to microbial invasion or tissue damage. It serves to eliminate pathogens and facilitate tissue repair, thereby playing a crucial role in maintaining physiological homeostasis. However, excessive or persistent inflammation may contribute to the onset and progression of various chronic diseases, including rheumatoid arthritis, cardiovascular disorders, and cancer. Therefore, the development of safe and effective anti-inflammatory agents is of great clinical importance for the prevention and treatment of these conditions.

Studies have shown that mulberry extract can promote the polarization of macrophages toward an anti-inflammatory phenotype by upregulating M2 macrophage surface markers

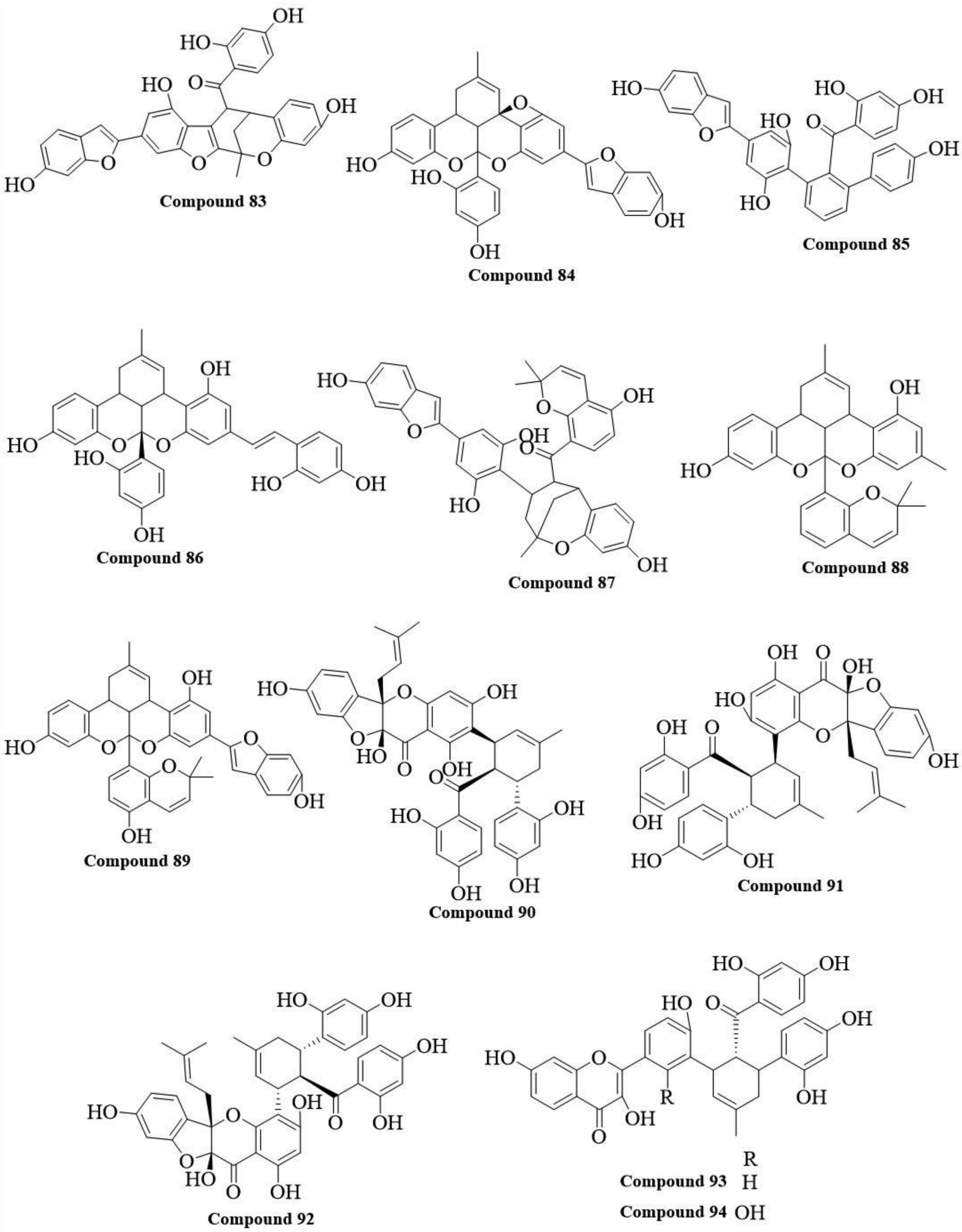


FIGURE 1 | (Continued)

such as CD163 and arginase-1 (ARG1), thereby alleviating inflammation in rat adipose tissue [119]. Resveratrol, one of the major bioactive compounds in mulberries, has been shown to significantly inhibit the expression of the pro-inflammatory enzyme cyclooxygenase-2 (COX-2) in mice, contributing to its anti-inflammatory effects [120]. Colitis is a chronic, nonspecific inflammation of the colon and rectum that may increase the risk of developing colon cancer. Anthocyanidins derived from mulberries exhibit significant protective effects against dextran sulfate sodium (DSS)–induced ulcerative colitis. The underlying mechanisms involve the inhibition of colonic oxidative stress, restoration of tight junction protein expression, and modulation of gut microbiota composition [121]. In addition, oxidized resveratrol isolated from mulberry branches exerts anticolitis effects via multiple mechanisms, including suppression of colonic mucosal cell proliferation, reduction of plasma levels of pro-inflammatory cytokines and nitric oxide (NO), downregulation of and inducible NO synthase (iNOS) gene expression, and decreased myeloperoxidase (MPO) activity [122]. Based on the above findings, the development of functional beverages or daily health products derived from mulberries or mulberry branches may offer

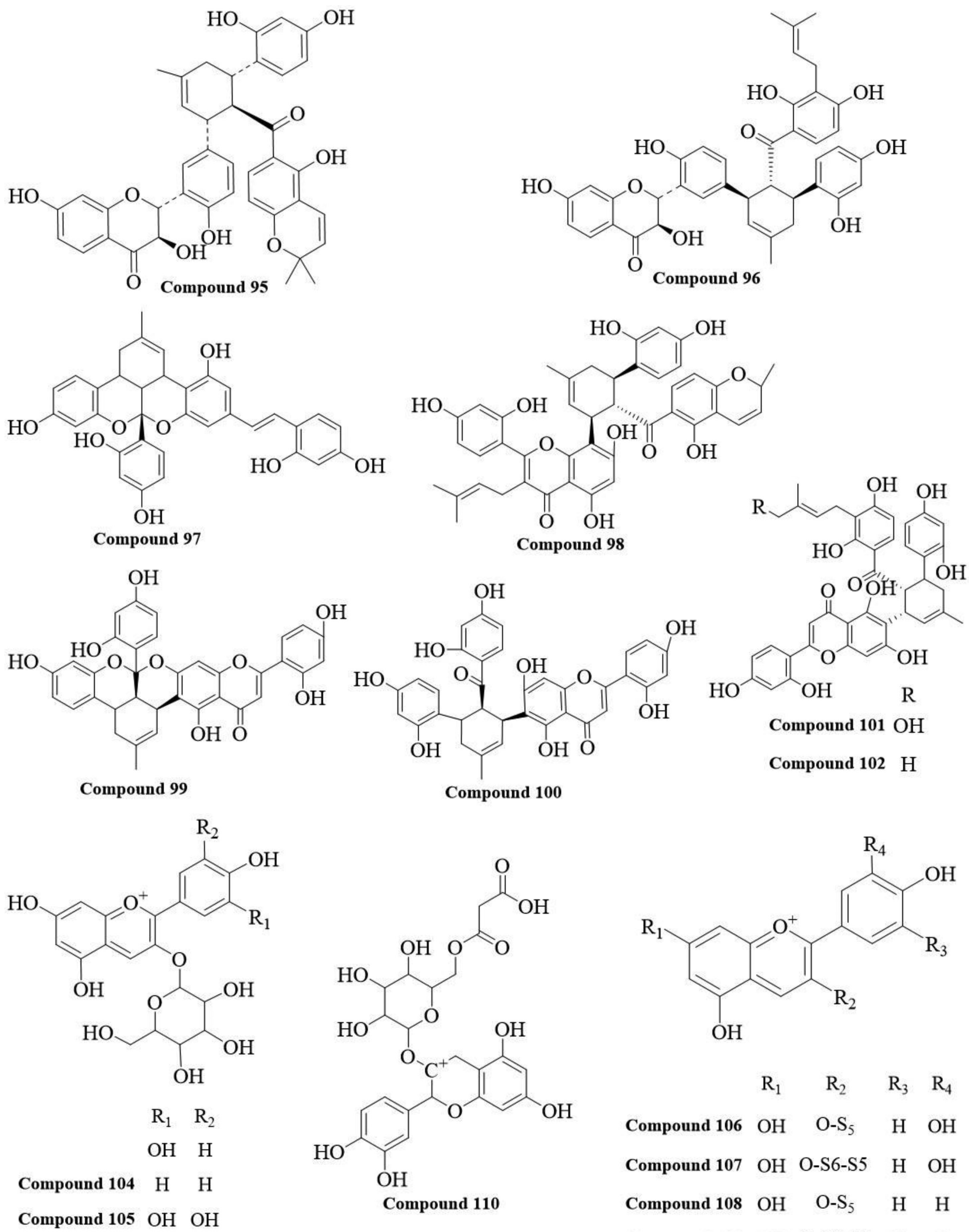


FIGURE 1 | (Continued)

promising strategies for the prevention and adjunctive treatment of chronic colitis.

Other parts of *M. alba* also exhibit significant anti-inflammatory activity. Extracts from mulberry leaves primarily exert their effects by inhibiting macrophage activation and reducing the production of pro-inflammatory mediators and cytokines [123]. Further mechanistic studies have demonstrated that neochlorogenic acid in mulberry leaves enhances superoxide dismutase (SOD) activity and downregulates pro-inflammatory factor expression, providing additional support for the multitarget anti-inflammatory potential of *M. alba*-derived constituents [124]. Additionally, extracts from the stem and root bark of *M. alba* have demonstrated considerable anti-inflammatory potential by significantly suppressing NO production and the mRNA expression of inducible iNOS in LPS-stimulated cells [125].

Notably, Sanghuang has demonstrated remarkable anti-inflammatory activity. In vitro studies have shown that Sanghuang extract inhibits the production of NO and prostaglandin E2 (PGE2) in a dose-dependent manner in LPS-stimulated

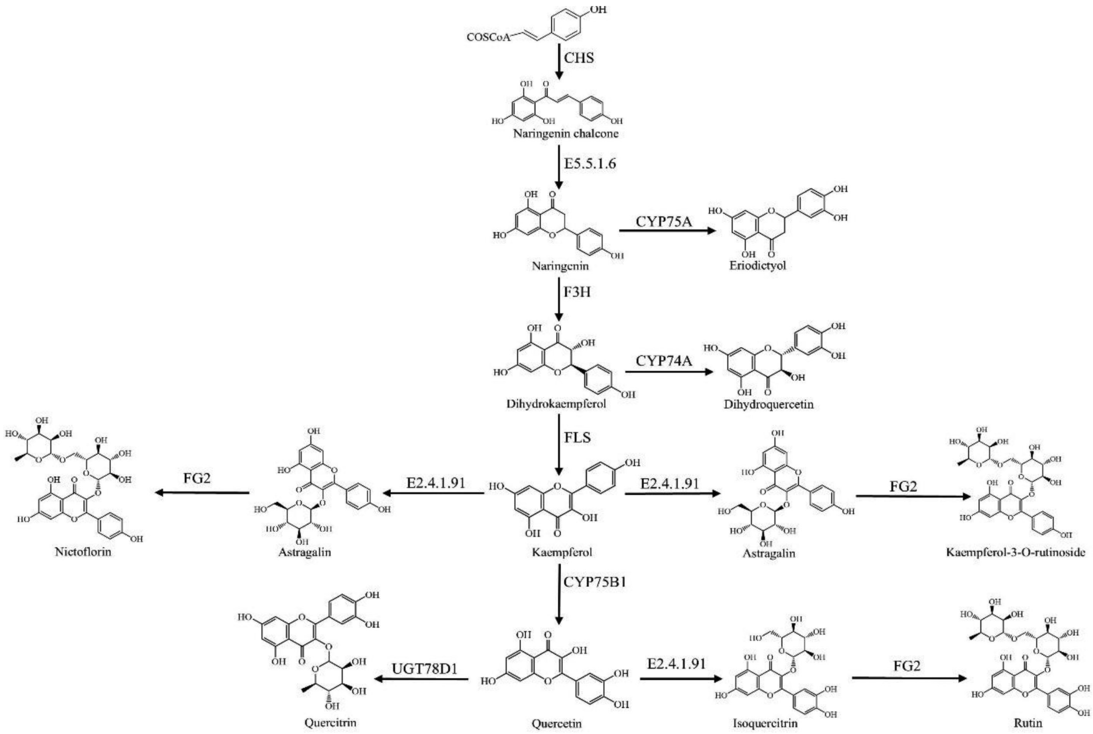


FIGURE 2 | The biosynthetic pathway of key flavonoid compounds in *Morus alba*.

RAW264.7 macrophages, while also suppressing the release of pro-inflammatory cytokines [126]. Hispolon, one of its key bioactive components, blocks inflammatory signaling cascades at the molecular level by inhibiting the activation of nuclear factor-kappa B (NF-κB) and signal transducer and activator of transcription 3 (STAT3) [127].

Given the significant anti-inflammatory effects and favorable bioavailability of extracts from various parts of *M. alba*, the development of diverse pharmaceutical formulations may not only enhance patient compliance but also improve convenience for daily use, thereby offering novel therapeutic options for the clinical management of inflammatory diseases.

4.4 | Anticancer Activity

Cancer remains one of the leading causes of death globally, with rising mortality and recurrence rates in recent years [128]. Conventional cancer treatments, such as chemotherapy, surgery, and hormone therapy, are often associated with severe adverse effects [129, 130]. In response to these limitations, natural products are increasingly being investigated as promising sources of anticancer agents, with the goal of reducing the side effects of conventional drugs and enhancing therapeutic efficacy. Recent studies have demonstrated that various medicinal parts of *M. alba* exhibit notable anticancer activities, characterized by low toxicity and favorable cost-effectiveness, thus earning increasing acceptance among patients [131, 132].

The antiproliferative effects of methanolic extracts from mulberry leaves on HCT-15 and MCF-7 cancer cell lines were assessed using the MTT assay. The results showed that the extracts significantly inhibited the proliferation of both cell lines. This effect is likely mediated through the activation of caspase-3, leading to apoptosis in HCT-15 and MCF-7 cells [133]. Moreover, DNJ, a bioactive compound in mulberry leaves, suppressed the progression of murine colorectal cancer in a dose-dependent manner by upregulating Bax mRNA expression and downregulating Bcl-2 mRNA levels in tumor cells [134].

The extract of Sangbaipi induces growth arrest and apoptosis in human colorectal SW480 cells via GSK3β- and ROS-dependent activation of ATF3 [135]. Additionally, the isolated compound morusin exhibits dose- and time-dependent inhibitory effects on the viability of A549 and NCI-H292 human lung cancer cells. The underlying mechanism involves multiple steps, including loss of mitochondrial membrane potential (MMP), release of cytochrome c into the cytosol, and cleavage of caspase-9 and calpain I, ultimately leading to apoptosis [131].

The bioactive compounds present in Sanghuang have demonstrated promising potential in the prevention and treatment of various cancers. Research has demonstrated that Sanghuang polysaccharides can inhibit the growth of HepG2 human liver cancer cells, with the inhibitory effect increasing with higher drug concentrations and longer treatment durations [136]. Additionally, Sanghuang ethanolic extracts have demonstrated significant inhibitory effects on lung cancer cell proliferation both in vitro and in vivo. The underlying mechanisms involve decreasing the ratio of proapoptotic Bcl-2/Bax proteins, inhibiting the expression of proliferative proteins cyclin D1 and CDK4, and downregulating STAT3 expression. Through these pathways, Sanghuang ethanolic extracts can dose-dependently suppress cancer cell viability, migration, and induce apoptosis [137, 138].

TABLE 3 | The detailed information on alkaloids compounds in different parts of *Morus alba*.

No.	Classification	Components	Sources	Reference
111	Alkaloids	1-Deoxynojirimycin (DNJ)	Mori Ramulus Mori Folium	[89]
112		Fagomine	Mori Ramulus Mori Folium	[89, 90]
113		*N*-Methyl-1-deoxynojirimycin	Mori Cortex (root bark)	[91]
114		3-Epi-fagomine	Mori Folium	[92]
115		Isofagomine	Mori Folium	[90]
116		2-*O*-α-D-Gal-DNJ	Mori Folium	[90]
117		4-*O*-β-D-Glc-Fagomine	Mori Folium	[90]
118		1,4-Dideoxy-1,4-imino-D-arabinitol	Mori Ramulus Mori Folium	[89, 91]
119		4-[Formyl-5-(hydroxymethyl)-1*H*-pyrrol-1-yl]butanoate	Mori Fructus	[92]
120		4-[Formyl 5-(methoxymethyl)-1*H*-pyrrol-1-yl]butanoic acid	Mori Fructus	[92]
121		Methyl 2-[2-formyl-5-(methoxymethyl)-1*H*-pyrrole-1-yl]propanoate	Mori Fructus	[92]
122		Morrole B	Mori Fructus	[92]
123		Morrole C	Mori Fructus	[92]
124		Morrole D	Mori Fructus	[92]
125		Methyl 2-[2-formyl-5-(methoxymethyl)-1*H*-pyrrol-1-yl]-3-(4-hydroxyphenyl)propanoate	Mori Fructus	[92]
126		2-(5-Hydroxymethyl-2-formylpyrrol-1-yl)propionic acid lactone	Mori Fructus	[92]
127		2-(5-Hydroxymethyl-2-formylpyrrol-1-yl)isovaleric acid lactone	Mori Fructus	[92]
128		2-(5-Hydroxymethyl-2-formylpyrrol-1-yl)isocaproic acid lactone	Mori Fructus	[92]
129		2-[2-Formyl-5-hydroxymethyl-1-pyrrolyl]-3-methylpentanoic acid lactone	Mori Fructus	[92]
130		2-(5′-Hydroxymethyl-2′-formylpyrrol-1′-yl)-3-phenyl-propionic acid lactone	Mori Fructus	[92]
131		2-(5′-Hydroxymethyl-2′-formylpyrrol-1′-yl)-3-(4-hydroxyphenyl)propionic acid lactone	Mori Fructus	[92]
132		2-(5-Hydroxymethyl-2′,5′-dioxo-2′,3′,4′,5′-tetrahydrobipyrrole)carbaldehyde	Mori Fructus	[92]
133		Morrole E	Mori Fructus	[92]
134		Morrole F	Mori Fructus	[92]
135		5-Hydroxymethyl-1*H*-pyrrole-2-carboxaldehyde	Mori Fructus	[92]
136		2-Formyl-1*H*-pyrrole-1-butanoic acid	Mori Fructus	[92]
137		2-Formyl-5-hydroxymethyl-1*H*-pyrrole-1-butanoic acid	Mori Fructus	[92]
138		2-Formyl-5-methoxymethyl-1*H*-pyrrole-1-butanoic acid	Mori Fructus	[92]
139		Morrole A	Mori Fructus	[92]
140		Betaine	Mori Folium	[91]

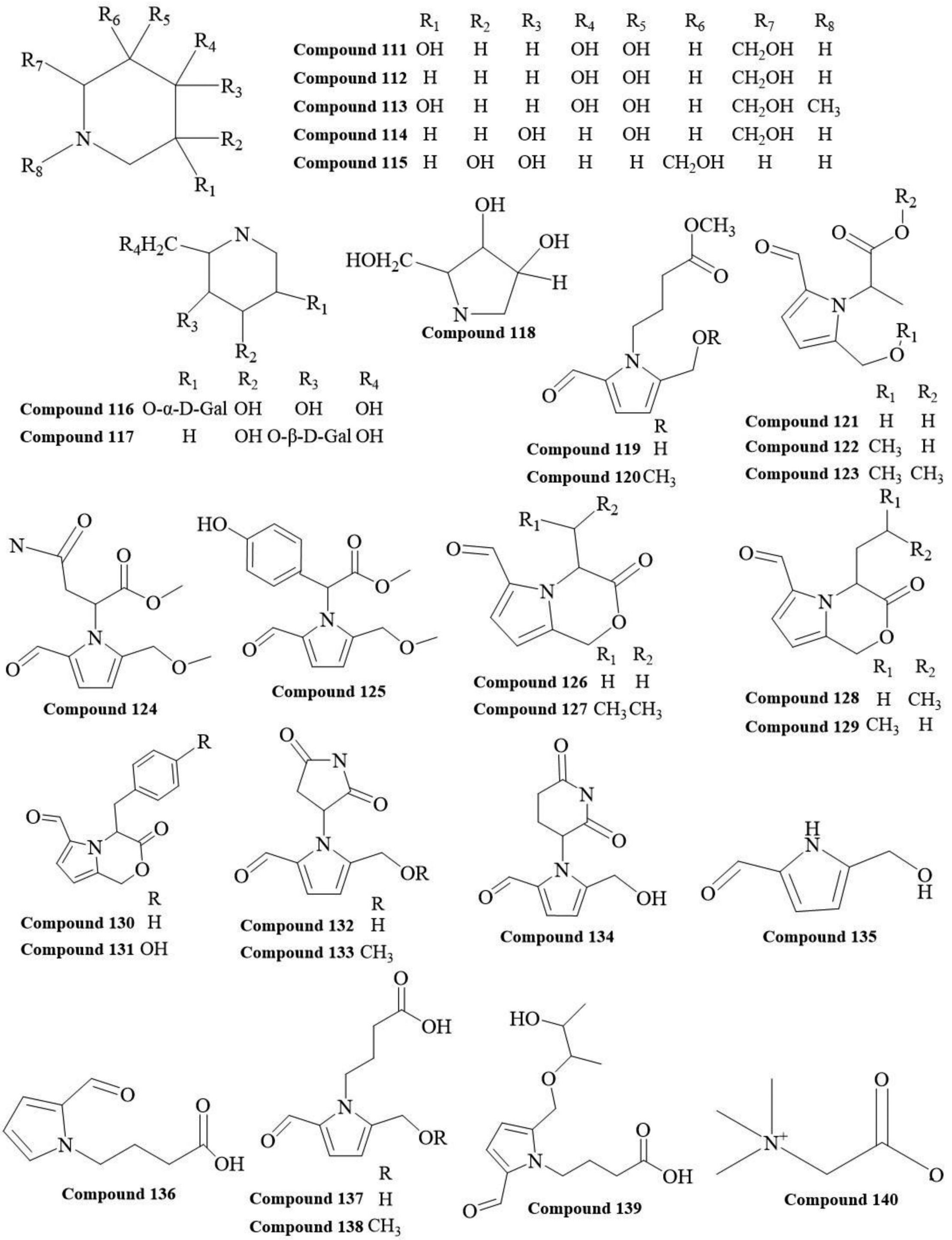


FIGURE 3 | The alkaloids chemical structures in different parts of *Morus alba*.

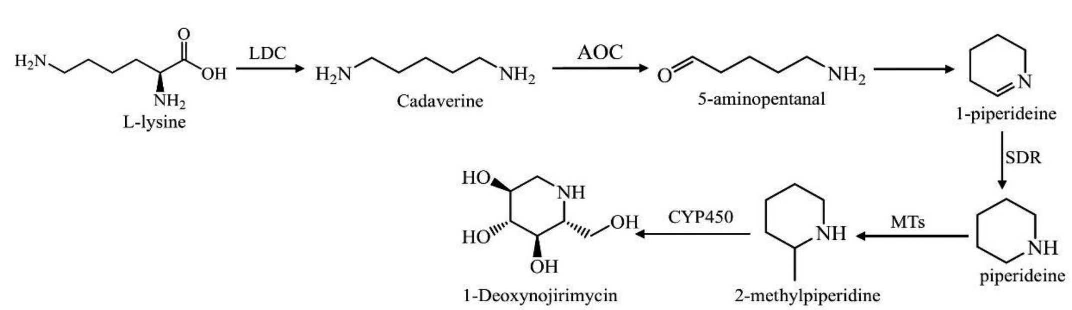


FIGURE 4 | The biosynthetic pathway of key alkaloid compounds in *Morus alba*.

TABLE 4 | The detailed information on terpenoid (a), coumarins (b), benzofurans (c) and stilbene (d) compounds in different parts of *Morus alba*.

No.	Classification	Components	Sources	Reference
141	Terpenoids	Lupeol	Mori Cortex (root bark)	[93]
142		Betulin	Mori Cortex (root bark)	[93]
143		Betulinic acid	Mori Cortex (root bark)	[93]
144		Butyrospermol	Mori Cortex (root bark)	[93]
145		Butyrospermol acetate	Mori Cortex (root bark)	[93]
146		α-Amyrin	Mori Cortex (root bark)	[93]
147		α-Amyrin acetate	Mori Cortex (root bark)	[93]
148		Ursolic acid	Mori Cortex (root bark)	[93]
149		Camelliols C	Mori Cortex (root bark)	[93]
150		Uvaol	Mori Cortex (root bark)	[55]
151		β-Sitosterol	Mori Cortex (root bark)	[55]
152		2-Methylpyrazine	Mori Folium	[94]
153		2,5-Dimethylpyrazine	Mori Folium	[94]
154		2,6-Dimethylpyrazine	Mori Folium	[94]
155		2-Ethylpyrazine	Mori Folium	[94]
156		2-Ethyl-3-methylpyrazine	Mori Folium	[94]
157		2-Ethyl-5-methylpyrazine	Mori Folium	[94]
158		Dihydroactinidiolide	Mori Folium	[94]
159		Loliolide	Mori Folium	[94]
160		(*E*)-Geranyl acetone	Mori Folium	[94]
161		(*E*)-α-Ionone	Mori Folium	[94]
162		(*E*)-β-Ionone	Mori Folium	[94]
163		β-Ionone-5,6-epoxide	Mori Folium	[94]
164	Coumarins	Scopolin	Mori Folium	[61]
165		Skimmin	Mori Folium	[61]
166		5,7-Dihydroxycoumarin-7-*O*-β-D-glucopyranoside	Mori Cortex (root bark)	[55]
167		5,7-Dihydroxycoumarin-7-*O*-β-D-apiofuranosyl-(1→6)-*O*-β-D-glucopyranoside	Mori Cortex (root bark)	[55]
168		7-Hydroxy coumarin	Mori Fructus Mori Folium	[60, 62]
169		Eseuletin	Mori Fructus	[60]
170		7-Hydroxy-6-methoxy coumarin	Mori Fructus Mori Folium	[60, 62]
171	Benzofurans	Mulberrofuran Y	Mori Cortex (root bark)	[54–56]
172		Moracin C	Mori Cortex (root bark) Mori Folium	[54, 55, 58]
173		Mulberroside F	Mori Cortex (root bark)	[95]
174		Moracin M	Mori Cortex (root bark) Mori Folium Mori Ramulus	[56, 62, 95]

(Continues)

No.	Classification	Components	Sources	Reference
175		Moracin J	Mori Ramulus	[57]
176		Mulberrofuran L	Mori Cortex (root bark)	[55, 56]
177		Moracin R	Mori Cortex (root bark)	[55]
178		Moracin Y	Mori Folium	[96]
179		Artoindonesianin O	Mori Cortex (root bark)	[57]
180		Alabafuran A	Mori Cortex (root bark)	[57]
181		Moracin B	Mori Cortex (root bark) Mori Ramulus	[56, 57]
182		Mulberrofuran A	Mori Cortex (root bark)	[56]
183		Moracin S	Mori Cortex (root bark)	[56]
184		Moracin N	Mori Cortex (root bark) Mori Folium	[55, 56, 58]
185		Mulberrofuran W	Mori Cortex (root bark)	[56]
186		Morusalfuran B	Mori Cortex (root bark)	[56]
187		Morusalfuran C	Mori Cortex (root bark)	[56]
188		Morusalfuran D	Mori Cortex (root bark)	[56]
189		Morusalfuran E	Mori Cortex (root bark)	[56]
190		Sanggenofuran A	Mori Cortex (root bark)	[56]
191		Mulberrofuran D	Mori Cortex (root bark)	[56]
192		Morusalfuran F	Mori Cortex (root bark)	[56]
193		Moracin E	Mori Cortex (root bark)	[56]
194		Morusalfuran A	Mori Cortex (root bark)	[56]
195		Morusalfuran G	Mori Cortex (root bark)	[56]
196		Moracin D	Mori Folium Mori Cortex (root bark) Mori Ramulus	[57, 58]
197		Moracin W	Mori Folium	[96]
198		Moracin X	Mori Folium	[96]
199		Moracin P	Mori Cortex (root bark)	[55]
200		Moracin V	Mori Folium	[96]
201		Moracin O	Mori Cortex (root bark)	[55]
202		Mulberrofuran D2	Mori Cortex (root bark)	[56]
203		Wittifurans P	Mori Ramulus	[75]
204		Wittifurans R	Mori Ramulus	[75]
205		Mulberrofuran H	Mori Cortex (root bark)	[54]
206	Stilbene	Mulberroside A	Mori Ramulus Mori Cortex (root bark)	[97]
207		Mulberroside F	Mori Ramulus	[97]
208		*trans*-Resveratrol-4′-*O*-β-D-glucopyranoside	Mori Ramulus	[97]

(Continues)

TABLE 4 | (Continued)

No.	Classification	Components	Sources	Reference
209		*trans*-Resveratrol-3′-*O*-β-D-glucopyranoside	Mori Ramulus	[97]
210		Oxyresveratrol-2-*O*-β-D-glucopyranoside	Mori Cortex (root bark)	[55]
211		Resveratrol	Mori Ramulus	[98]
212		Oxyresveratrol	Mori Ramulus	[57, 99]

In summary, various bioactive compounds derived from different parts of the *M. alba* possess anticancer potential and exert their effects through multiple molecular mechanisms. These research findings lay the foundation for further development and clinical applications of *M. alba* bioactive compounds in cancer therapy. *M. alba* bioactive compounds hold promise as potential anticancer drug candidates.

4.5 | Hepatoprotective Activities

Alcoholic liver disease (ALD) is a condition caused by excessive alcohol consumption, characterized by hepatic steatosis, cirrhosis, and the potential development of hepatocellular carcinoma. Alcohol intake leads to increased inflammation and oxidative stress, ultimately resulting in liver injury and the progression of steatosis, cirrhosis, and liver cancer [139–141]. Mulberry leaf extract (MLE) has been shown to reverse alcohol-induced liver injury in a dose-dependent manner. The underlying mechanisms involve increasing acetaldehyde (ACE) levels, decreasing aldehyde dehydrogenase (ALDH) activity, inhibiting leukocyte infiltration and production of pro-inflammatory factors, restoring alcohol-induced hepatocyte apoptosis. Consequently, MLE exerts its protective effects on liver cells through multiple pathways, thereby reversing the toxic effects of alcohol. These findings suggest that MLE has potential therapeutic value in treating ALD [142, 143].

Research has demonstrated that the alcoholic extract of mulberry fruits can inhibit the phosphorylation of the mitogen-activated protein kinase (MAPK) pathway, including the expression of p38, c-Jun N-terminal kinase (JNK), extracellular signal–regulated kinase (ERK), and caspase-3 proteins, thereby exerting protective effects against alcohol-induced injury [144]. Another study found that the aqueous extract of mulberry fruits can prevent alcohol-induced liver injury through activating AMP-activated protein kinase (AMPK), reducing the synthesis of triglycerides (TGs) and cholesterol to inhibit lipid accumulation in the liver. Simultaneously, by activating the AMPK and peroxisome proliferator–activated receptor-α (PPAR-α) signaling pathways, it enhances fatty acid oxidation in the liver and increases the expression of the fatty acid transporter carnitine palmitoyltransferase-1 (CPT-1) and the microsomal TG transfer protein (MTP) [145]. These findings suggest that different parts of the *M. alba*, including the fruits, possess potential benefits for preventing and treating alcohol-induced liver injury.

In summary, mulberry extracts have demonstrated potential therapeutic effects in treating ALD. These findings provide important theoretical and experimental foundations for developing novel treatment strategies.

4.6 | Other Health Benefits

In addition to the aforementioned effects, studies have revealed that certain phytochemical components present in mulberry also exhibit significant pharmacological activities, including anti-obesity [146], antidepressant [147], antihypertensive [148], and cardioprotective properties [149]. However, the specific mechanisms of action and molecular pathways underlying these diverse bioactivities warrant further in-depth investigation and elucidation.

5 | Products Development From Various Parts of the *M. alba*

5.1 | Functional Beverage

Mulberry wine, as one of the value-added products derived from mulberry, possesses potential health-promoting properties and a distinctive organoleptic profile. In recent years, it has attracted increasing attention in the functional beverage market. The selection of appropriate yeast strains plays a critical role in determining the final quality of mulberry wine. Studies have shown that among five evaluated commercial yeast strains, the L13 strain significantly enhanced the sensory quality and overall flavor of mulberry wine, thereby facilitating product development [150]. Additionally, mixed fermentation can effectively increase the content of functional compounds, thereby enhancing the sensory experience and functional characteristics of the product [151]. Non-Saccharomyces yeasts have also exhibited potential for producing low-alcohol mulberry wines by reducing alcohol content through single or mixed inoculation methods [152]. In addition to improvements in sensory quality, recent studies have shown that mulberry wine is increasingly recognized by younger consumers and the high-end beverage market due to its natural origin, low sugar content, and functional attributes. Moreover, composite processing with different raw materials has enabled the development of diversified flavor profiles. For instance, a functional mulberry wine formulated by blending 2% mulberry leaf powder with mulberry juice exhibited excellent antioxidant capacity and favorable consumer acceptance [153]. Kombucha, a traditional fermented beverage made from mulberries and other ingredients with a history spanning thousands of years in China, stands out in terms of color, aroma, and taste and also offers health benefits, making it increasingly popular in western and middle eastern regions [154]. Mulberry-based fermented beverages exhibit significant market potential and feasibility for large-scale commercialization. To fully realize this potential, further comprehensive studies are warranted to support product diversification, enhance industrial supply chain integration, and establish standardized production protocols.

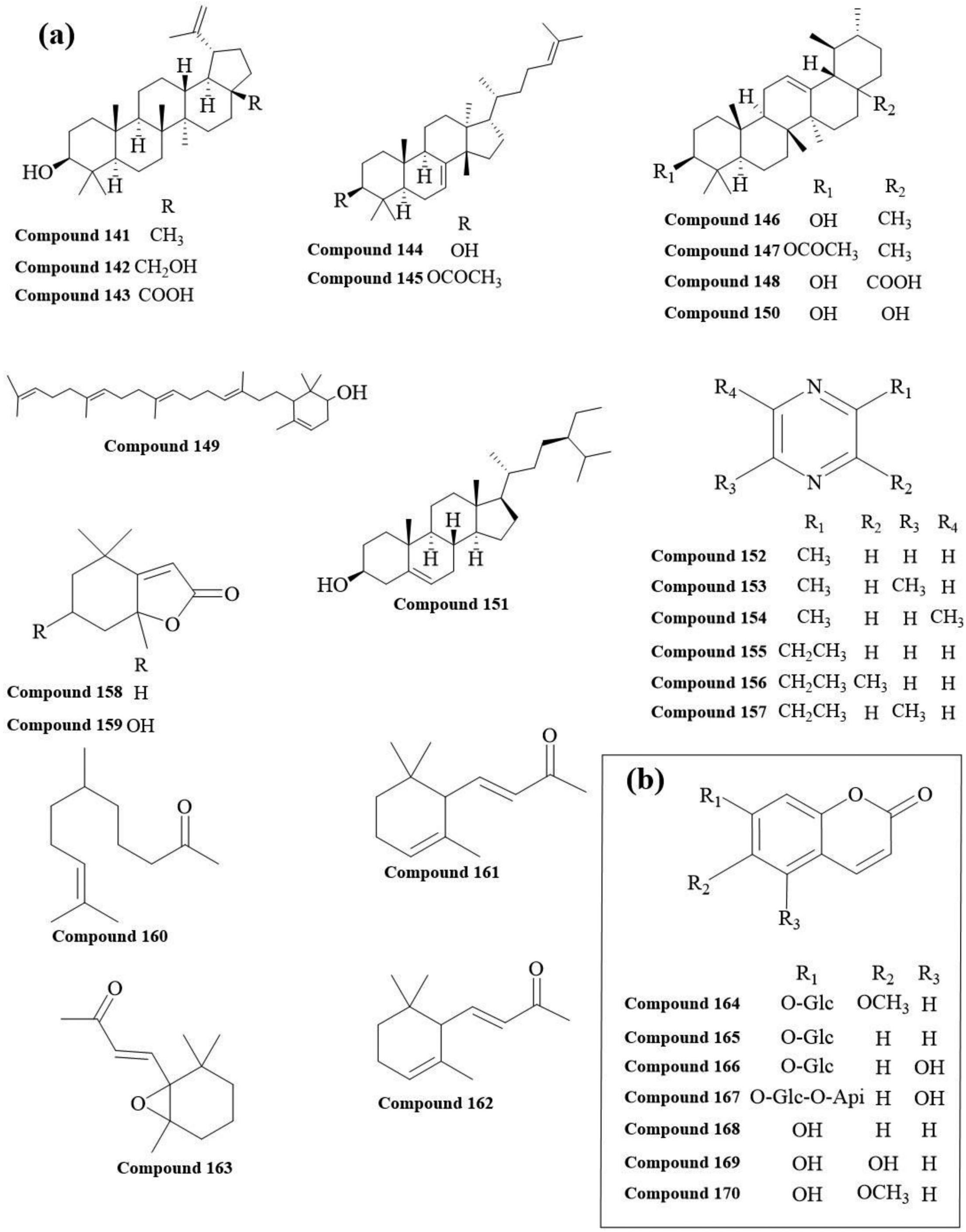
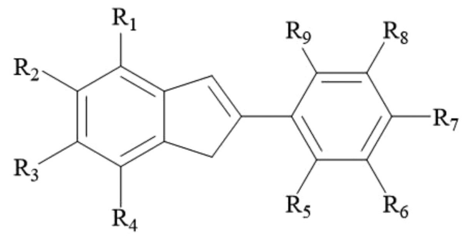


FIGURE 5 | The chemical structures of terpenoids (a), coumarins (b), benzofurans (c), and stilbene (d) in different parts of *Morus alba*.

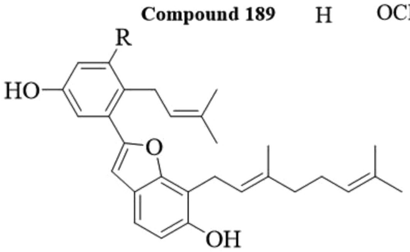
5.2 | Functional Foods

The incorporation of 4% mulberry leaf powder into cassava-based food products did not adversely affect their organoleptic attributes; instead, it exhibited enhanced antioxidant potential and improved nutritional profile, showcasing the application potential of mulberry leaf powder in producing high-quality food products [155]. The combination of mulberry fruit juice with low-fat, low-sugar ice cream not only enhanced the textural complexity but also increased the nutritional value, rendering it suitable for promotion to the general public [156]. Mulberry fruit powder not only extends shelf life but can also be incorporated into various food products. Notably, the jelly made from mulberry fruit powder is not only nutrient rich but also reduces the risk of atherosclerosis, meeting the demand for functional foods [157]. The low-sugar mulberry fruit jam produced through fermentation not only retains the aroma of mulberry fruits but also possesses the unique flavor and fragrance imparted by fermentation, with a significant increase in organic acids and aroma compounds after fermentation [158]. The dairy sector also demonstrates promising application potential. The incorporation of mulberry powder into yogurt not only enhances its

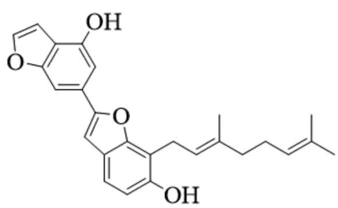
(c)



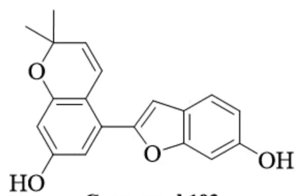
	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9
Compound 171	geranyl	OCH_3	OH	H	H	OH	H	OH	H
Compound 172	H	H	OH	H	H	OH	prenyl	OH	H
Compound 173	H	H	O-S_5	H	H	O-S_5	H	OH	H
Compound 174	H	H	OH	H	H	OH	H	OH	H
Compound 175	H	OH	OCH_3	H	H	OH	H	OH	H
Compound 176	H	H	OH	geranyl	H	OH	H	OH	H
Compound 177	H	S_7	OH	H	H	OH	H	OH	H
Compound 178	H	CHO	OH	H	H	OH	H	OH	H
Compound 179	H	H	OH	H	H	OCH_3	prenyl	OH	H
Compound 180	H	H	OH	H	geranyl	OH	H	OH	H
Compound 181	H	OH	OCH_3	H	H	OCH_3	H	OH	H
Compound 182	H	H	OH	H	geranyl	OCH_3	H	OH	H
Compound 183	H	H	OH	prenyl	H	OH	H	OH	H
Compound 184	H	prenyl	OH	H	H	OH	H	OH	H
Compound 185	H	H	OH	H	S_9	OH	H	OH	H
Compound 186	H	OH	H	H	prenyl	OH	H	OH	S_9
Compound 187	H	OH	H	H	H	OH	H	OCH_3	S_9
Compound 188	prenyl	OH	H	H	H	OH	prenyl	OH	H
Compound 189	H	OCH_3	H	H	H	OCH_3	H	OH	H



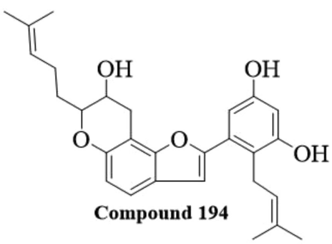
	R
Compound 190	OCH_3
Compound 191	OH



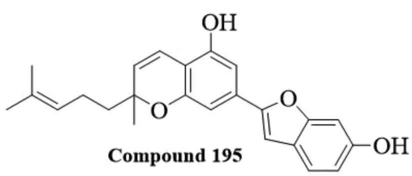
Compound 192



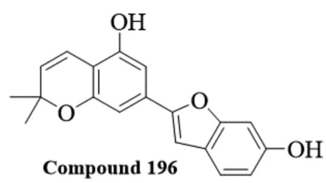
Compound 193



Compound 194



Compound 195



Compound 196

FIGURE 5 | (Continued)

color, aroma, and flavor profile but also significantly improves its nutritional composition, thereby increasing consumer acceptability [159]. Furthermore, a variety of composite yogurt products formulated with mulberries and other ingredients—such as goji berries or taro—have emerged, offering enhanced nutritional value and health benefits compared to conventional yogurts [160, 161].

Mulberry leaves and fruits are traditional agricultural resources in China, featuring convenient access to raw materials and controllable production costs. They hold great potential for the development of functional foods.

5.3 | Food Freshness Indicators and Preservatives

Exploiting the color-changing property of anthocyanins from mulberries under varying pH conditions [162], a natural and nontoxic pH indicator has been developed for the fabrication of intelligent packaging materials capable of indicating food

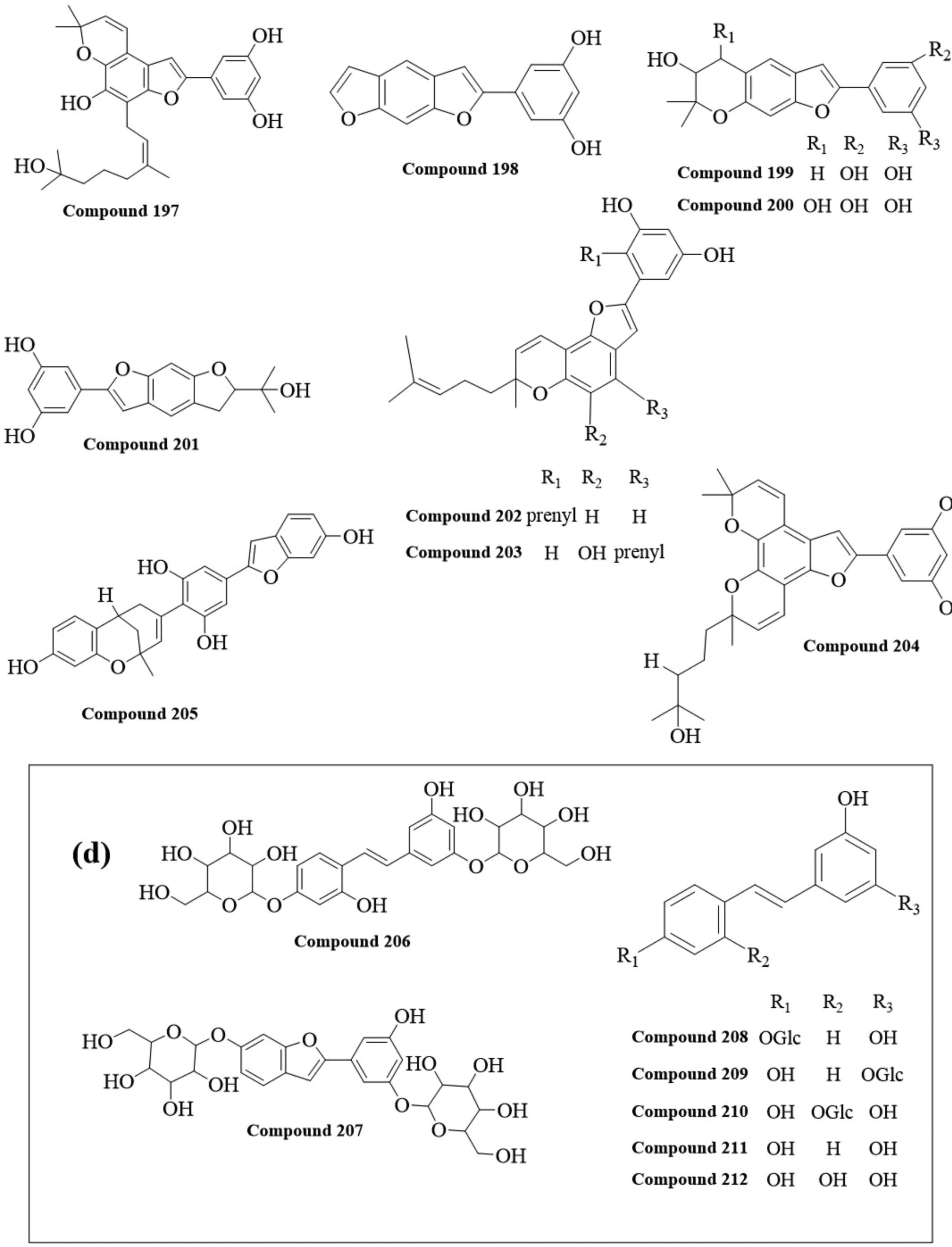


FIGURE 5 | (Continued)

freshness. Upon food spoilage, the pH shift in the food induces a corresponding color transition in the pH indicator, allowing real-time tracking of food freshness [163].

Zheng et al. [164] successfully fabricated an intelligent pH-responsive membrane by incorporating collagen, chitosan, zinc oxide nanoparticles, and mulberry anthocyanins. Compared to traditional collagen-chitosan membranes, this novel smart membrane not only demonstrated superior mechanical strength and antioxidant properties but also exhibited color changes in response to pH variations in the buffer solution, showing excellent potential for freshness detection in pork. Similarly, Zeng et al. [165] developed a pH-sensitive indicator membrane based on a gelatin/polyvinyl alcohol matrix and anthocyanin extracts from mulberry pomace to monitor fish spoilage. Additionally, a bilayer colorimetric indicator membrane containing mulberry anthocyanins showed superior applicability for freshness indication of mitten crabs [166].

Beyond their role as food freshness indicators, mulberry anthocyanins also impart antioxidant and natural antimicrobial properties. These properties mitigate food oxidation and inhibit

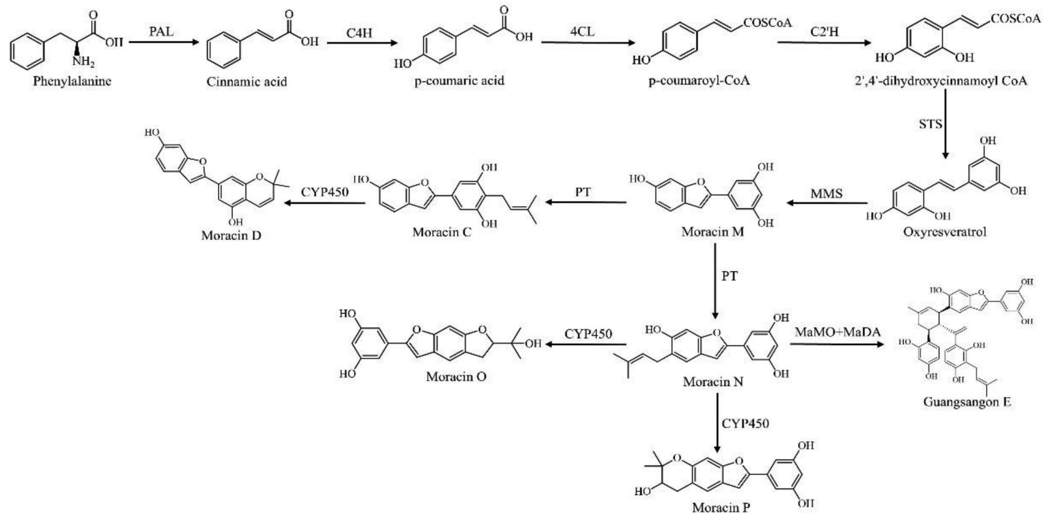


FIGURE 6 | The biosynthetic pathway of moracins in *Morus alba*.

microbial growth, thereby extending food shelf life. For instance, Turan and Şimşek investigated the preservative effects of incorporating freeze-dried black mulberry water extract powder into beef patties. The results revealed that the powder not only maintained color stability in the patties but also inhibited lipid and protein oxidation and suppressed spoilage microorganism growth, thus extending shelf life [167]. Furthermore, edible composite films incorporating MLEs and 1-DNJ into a pectin matrix demonstrated potent antioxidant and antimicrobial activities [168].

These studies collectively demonstrate the promising applications of mulberry anthocyanins not only in intelligent food packaging and freshness monitoring but also as natural preservatives in the fruit and vegetable industry, contributing to the extended shelf life of perishable products. Despite the initial integration of natural pigments into smart packaging materials, their broader adoption remains limited by the high cost of extraction and the insufficient environmental stability of packaging films under complex conditions. Therefore, future research should prioritize the systematic optimization of anthocyanin extraction techniques and material formulations to improve their functional stability and environmental adaptability, thereby further expanding their potential in smart packaging applications.

5.4 | Chinese Patent Medicine and Chinese Medicine Preparations

Herbal medicines, both in China and globally, have been extensively documented for their potential in preventing and treating various diseases, gaining widespread attention [169]. These herbal formulations are the result of thousands of years of accumulated experience by ancient Chinese physicians. Herbal medicines derived from different parts of the *M. alba*, whether used individually or formulated into herbal compounds or proprietary Chinese medicines, hold immense potential for treating and preventing a wide range of diseases. Today, TCM formulations have been modernized into various dosage forms, including pills, oral solutions, and others [170].

Sangzhi Alkaloid Tablets, derived from mulberry branches, are the first TCM approved for the treatment of DM in China and the first original natural hypoglycemic drug [171]. These tablets, which have a clear therapeutic target, well-defined chemical composition, controllable quality, and significant blood glucose-lowering effects, help overcome skepticism regarding the quality control and efficacy of traditional herbal medicines in the international market [172]. A compound formulation consisting of *Taxilli Herba*, mulberry fruit, and branch has been utilized in treating diabetes-induced arthropathy and peripheral neuropathy [173]. Academician Boli Zhang has extensively employed the Tangzhiqing formula, comprising mulberry leaf, lotus leaf, and other herbs, in the clinical management of hyperlipidemia and DM. This formula primarily acts by modulating adipocyte differentiation and insulin action, alleviating hyperinsulinemia and obesity [174]. In addition to compound formulations, proprietary Chinese medicines derived from single herbs, such as Sangzhi Granules, have demonstrated efficacy in reducing fasting and postprandial blood glucose levels in diabetic patients, while also improving hyperlipidemia [175].

Studies have demonstrated that for patients with dry eye syndrome, combining conventional Western medicine with the oral administration of Sang Bai Pi Decoction yields significantly superior therapeutic effects compared to conventional Western medicine alone. This indicates that Sang Bai Pi Decoction plays a positive role in the treatment of dry eye syndrome and can serve as a complementary therapy to conventional Western medicine [176]. Yu et al.'s research has also confirmed the efficacy and feasibility of Sang Bai Pi Decoction in the treatment of dry eye syndrome [[177].

5.5 | Other Product Developments Related to Various Parts of the *M. alba*

The *M. alba* has extensive applications beyond the food industry, extending into diverse fields such as healthcare, cosmetics, textiles, and packaging materials. Figure 7 illustrates the various functional products developed from different parts of the

FIGURE 7 | Various functional products have been developed from different parts of *Morus alba*.

M. alba. In the cosmetics industry, a cream formulated by incorporating 4% concentrated mulberry extract into the aqueous phase of a water-in-oil (W/O) emulsion has been shown to exert significant skin-care effects. This formulation markedly reduced skin melanin content without causing any irritation. Moreover, relevant studies have demonstrated that it improves the stability of active ingredients and enhances their transdermal delivery, thereby significantly improving skin-care efficacy [178]. In addition, a facial gel mask formulated with mulberry extract, 7% polyvinyl alcohol (PVA), and 2.5% hydroxypropyl methylcellulose (HPMC) has demonstrated significant efficacy in improving common skin conditions such as acne, while also exhibiting mildness and nonirritating properties [179]. Mulberry branches, due to their high cellulose, lignin, and hemicellulose content, hold significant value in fields such as fiberboard production, wastewater treatment, and resource development. Compared to other nonwoody plant materials, mulberry branches have higher fiber content and lower xylan and ash content, making them a viable economic alternative to traditional wood and contributing to environmental protection [180]. Moreover, the anthocyanins found abundantly in mulberry fruits and leaves serve as natural colorants with excellent temporal stability [181], making them promising candidates for use in textile dyeing. Aqueous mulberry leaf solutions are employed to dye cotton fabrics and woolen yarns, yielding various shades. Cotton fabrics are dyed in yellow hues, while woolen yarns produce green, deep green, and yellow shades. Colorfastness tests—including washing, rubbing, and light exposure—revealed that cotton fabrics showed better dyeing performance than woolen yarns [182]. The natural dye derived from mulberry provides textiles with appealing colors while meeting contemporary demands for environmental protection and sustainable development.

6 | Conclusions and Prospects

M. alba is a valuable resource, with various parts containing a wide range of bioactive compounds that have potential medicinal value for human health. It serves not only as the cornerstone of the sericulture industry but also as an important medicinal plant. In recent years, with the advancement of modern science and technology and the continued deepening of research, the nutritional and therapeutic values of various bioactive compounds in *M. alba* have gained increasing attention. Bioactive compounds from different parts of *M. alba* have been progressively identified and isolated, laying the foundation for pharmacological studies. However, despite notable progress in related research, the biosynthetic pathways and pharmacological mechanisms of these bioactive compounds still face significant challenges.

Firstly, current research primarily focuses on the efficacy of individual compounds, whereas systematic investigations into the synergistic therapeutic effects of compound mixtures remain limited. Therefore, future research should integrate multidisciplinary approaches, such as pharmacokinetics, toxicology, and clinical trials, to provide more comprehensive theoretical support for the medicinal development of *M. alba*.

Secondly, compared to other parts of *M. alba*, research on *Mori Ramulus* and *Mori Cortex* remains limited, and their potential bioactive compounds and specific pharmacological effects have yet to be systematically elucidated. Current studies primarily focus on in vitro experiments, with a lack of functional evidence at the clinical level. Particularly, the pharmacological efficacy and safety of *Mori Cortex* constituents are still insufficiently supported by experimental data. Therefore, future research should focus on developing diversified extraction and structural characterization techniques to comprehensively and accurately reflect the bioactivity of various parts of *M. alba*. Additionally, with the rapid development of molecular docking and molecular dynamics simulation technologies, these computational methods have become widely applied in the study of pharmacological activity and mechanisms. Future studies could leverage these advanced technologies to screen active compounds from various parts of *M. alba* and explore their mechanisms of action, providing a solid theoretical foundation for clinical applications.

Thirdly, *Mori Fructus* is a perishable fruit with a short shelf life. Although several functional products have been developed, its full potential remains largely untapped. Future research should focus on the development of gene-edited probiotic yeasts derived from *Mori Fructus*, the design of transdermal sustained–release systems for its bioactive compounds, and the application of biofluorescent dyes in live-cell imaging and environmental pollutant detection.

Fourthly, *M. alba* has accumulated considerable genetic diversity throughout its evolutionary history, enabling the synthesis of structurally diverse and functionally versatile bioactive compounds. However, the chemical structures of many of these constituents remain incompletely characterized. Therefore, systematic analysis of *M. alba* resources is warranted to identify novel compounds with unique structures and to uncover previously unexplored metabolic pathways, thereby expanding our understanding of its phytochemical diversity and therapeutic potential.

In summary, future research should strengthen the in-depth investigation of various parts of *M. alba*, establish relevant standards, and promote the comprehensive and sustainable development and utilization of its resources.

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Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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